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Quantum Machine Learning for Brain Tumor Segmentation and Classification using MRI

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Dedication

I dedicate this work:

To my parents, for their unconditional love and the life-long sacrifices that have paved the way for my success. To my mother, my source of strength and comfort, whose wisdom and tireless support inspire me to persevere through every challenge. To my father, for his invaluable guidance and for instilling in me the core values of integrity and hard work. I am deeply grateful to them both for being my greatest models of determination and for encouraging me to pursue my dreams with confidence.

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Abstract

Brain tumor analysis from magnetic resonance imaging (MRI) is an important task in medical image analysis. This work proposes a MedSAM-based framework for automated brain tumor segmentation and classification using the BRISC dataset. MedSAM was adapted to the considered task through full fine-tuning and a multi-bounding-box prompting strategy. The proposed segmentation model achieved a Dice score of 95.06% and an IoU of 90.78% on the test set, demonstrating accurate tumor delineation. For classification, a classical multilayer perceptron (MLP) was compared with a variational quantum circuit. The selected quantum classifier, based on eight qubits, angle embedding, and three variational layers, achieved an accuracy of 85.40%, compared with 84.30% for the classical MLP, while using approximately 92% fewer trainable parameters. Overall, the results demonstrate the effectiveness of the proposed MedSAM-based framework for brain tumor segmentation and highlight the potential of compact quantum classification heads as an alternative to classical components.

Keywords: Brain Tumor Analysis; Magnetic Resonance Imaging; Medical Image Segmentation; MedSAM; Vision Foundation Models; Quantum Machine Learning.

Résumé

L'analyse des tumeurs cérébrales à partir d'images par résonance magnétique (IRM) constitue une tâche importante en analyse d'images médicales. Ce travail propose un cadre basé sur MedSAM pour la segmentation et la classification automatiques des tumeurs cérébrales à partir de la base de données BRISC. MedSAM a été adapté à la tâche considérée à travers un fine-tuning complet et une stratégie de prompting basée sur plusieurs boîtes englobantes. Le modèle de segmentation proposé a obtenu un score de Dice de 95.06% et un IoU de 90.78% sur l'ensemble de test, démontrant une délimitation précise des tumeurs. Pour la classification, un perceptron multicouche (MLP) classique a été comparé à un circuit quantique variationnel. Le classifieur quantique retenu, basé sur huit qubits, un encodage par angle et trois couches variationnelles, a atteint une précision de 85.40%, contre 84.30% pour le MLP classique, tout en utilisant environ 92% de paramètres entraînables en moins. Dans l'ensemble, les résultats démontrent l'efficacité du cadre proposé basé sur MedSAM pour la segmentation des tumeurs cérébrales et mettent en évidence le potentiel des têtes de classification quantiques compactes comme alternative aux composants classiques.

Mots-clés : Analyse des tumeurs cérébrales ; Imagerie par résonance magnétique ; Segmentation d'images médicales ; MedSAM ; Modèles de fondation en vision ; Apprentissage automatique quantique.

Contents

Abstract **III**

List of Figures **V**

List of Tables **VI**

List of Acronyms **VII**

General Introduction **1**

1 General Context and Problem Statement **3**
1.1 Introduction 4
1.2 Host Organization 4
1.2.1 The Mitacs Program 4
1.2.2 Université de Moncton 4
1.2.3 PRIME Laboratory 5
1.3 Brain Tumor Analysis: Clinical Background and Medical Context 5
1.4 Problem Statement 6
1.5 Project Planning 7
1.6 Conclusion 8

2 Related Work **9**
2.1 Introduction 10
2.2 Deep Learning Architectures for Brain Tumor Analysis 10
2.2.1 Brain Tumor Segmentation 10
2.2.2 Brain Tumor Classification 13
2.3 Brain Tumor MRI Datasets 14
2.4 Limitations of Existing Approaches 16
2.5 Conclusion 16

3 Proposed MedSAM-Based Methodology **17**
3.1 Introduction 18
3.2 Theoretical Background: the MedSAM Model 18
3.2.1 From SAM to MedSAM 18
3.2.2 Architectural Components 18
3.2.3 Prediction Mechanism 20
3.3 Proposed Framework and Data Processing Pipeline 20
3.3.1 Overview of the Proposed Framework 20
3.3.2 Adaptation of the BRISC Dataset 21
3.3.3 Preprocessing and Data Augmentation 22
3.3.4 Multi-Bounding-Box Prompting 23

3.3.5 Classification Branch . . . 25
3.4 Training Strategy . . . 26
3.4.1 Two-Stage Full Fine-Tuning . . . 26
3.4.2 Joint Objective . . . 27
3.4.3 Optimisation . . . 28
3.4.4 Training Configuration . . . 28
3.4.5 Computational Environment . . . 30
3.5 Evaluation Metrics . . . 31
3.5.1 Segmentation Metrics . . . 31
3.5.2 Classification Metrics . . . 32
3.6 Experimental Results . . . 32
3.6.1 Segmentation Performance . . . 32
3.6.2 Classification Performance . . . 33
3.6.3 Comparison with Retrained Architectures . . . 33
3.6.4 Comparison with the Published State of the Art . . . 34
3.7 Discussion . . . 35
3.8 Conclusion . . . 35

4 Quantum Integration and Experimental Analysis 36
4.1 Introduction . . . 37
4.2 Quantum Computing Background . . . 37
4.2.1 Fundamentals of Quantum Computing . . . 37
4.2.2 Variational Quantum Circuits . . . 40
4.3 Proposed Quantum Classification Head . . . 42
4.3.1 Placement of the Quantum Circuit . . . 42
4.3.2 Preparation of Classical Features . . . 43
4.3.3 Quantum Circuit Architecture . . . 43
4.3.4 Quantum-to-Classical Classification . . . 46
4.4 Experimental Setup . . . 46
4.5 Results . . . 47
4.5.1 Effect of the Data Embedding . . . 48
4.5.2 Effect of the Number of Qubits . . . 49
4.5.3 Effect of the Number of Variational Layers . . . 50
4.5.4 Comparison with the Classical MLP . . . 51
4.5.5 Parameter Efficiency and Computational Cost . . . 52
4.5.6 Qualitative Results . . . 52
4.6 Discussion . . . 53
4.7 Conclusion . . . 56

General Conclusion 57
Annex A – Complete Segmentation Results on BRISC 2025 . . . 59

Bibliography 60

List of Figures

1.1 Université de Moncton 4
1.2 Gantt chart showing the project timeline and task dependencies. 8

3.1 Structure of the ViT-B image encoder of MedSAM. 19
3.2 Architecture of the proposed MedSAM-based framework [1]. 21
3.3 Preprocessing, online data augmentation, and prompt generation pipeline. 23
3.4 Comparison of single- and multi-bounding-box prompting for a fragmented tumor. 24
3.5 MLP head for classification. 26

4.1 Proposed MedSAM architecture with quantum classification head for joint segmentation and classification. 42
4.2 Architecture of the proposed eight-qubit, three-layer variational quantum classification head employing data re-uploading and Z/ZZ measurements. 46
4.3 Training and validation curves of the classical MLP and the selected quantum classification head. 51
4.4 Qualitative results of the final framework on one representative case of each BRISC category. 53

List of Tables

2.1 Comparison of publicly available brain tumor MRI datasets. 15

3.1 Multi-bounding-box prompt generation. 23
3.2 Main training configuration of the proposed MedSAM-based framework. 29
3.3 Computational environment used for the experiments. 30
3.4 Segmentation performance of the proposed framework on the BRISC test set. . . 32
3.5 Classification performance of the MLP head on the BRISC test set. 33
3.6 Comparison of the proposed framework with representative segmentation architectures. 34

4.1 Quantum gates used in this work: unitary matrices and effect [2]. 39
4.2 Correspondence between the stages of a variational quantum circuit and those of a classical neural network. 41
4.3 Configuration used for the classification-head experiments. 48
4.4 Comparison between angle embedding and amplitude embedding using eight qubits. 48
4.5 Comparison between the classical MLP and quantum classification heads with different numbers of qubits. 49
4.6 Effect of the number of variational layers on classification performance, at eight qubits with angle embedding. 50
4.7 Comparison between the classical MLP and the best-performing eight-qubit quantum classification head. 51
4.8 Segmentation results obtained on the BRISC 2025 test set, ordered by decreasing Dice coefficient. Higher values indicate better performance. 59

List of Acronyms

BCD-MRI	Brain Cancer Dataset–MRI
BRISC	Brain Tumor Segmentation and Classification
BraTS	Brain Tumor Segmentation
FLAIR	Fluid-Attenuated Inversion Recovery
HGG	High-Grade Glioma
LGG	Low-Grade Glioma
MRI	Magnetic Resonance Imaging
MSD	Medical Segmentation Decathlon
T1ce	Contrast-Enhanced T1-Weighted
TCGA	The Cancer Genome Atlas
TCGA-LGG	The Cancer Genome Atlas Low-Grade Glioma
TCIA	The Cancer Imaging Archive
ABANet	Attention-Based Adaptive Network
BASNet	Boundary-Aware Segmentation Network
CNN	Convolutional Neural Network
MedSAM	Medical Segment Anything Model
SAM	Segment Anything Model
SegFormer	Segmentation Transformer
U-Net	U-shaped Network
ViT-B16	Vision Transformer Base with 16×16 Patches
YOLO	You Only Look Once
GELU	Gaussian Error Linear Unit
MHSA	Multi-Head Self-Attention
MLP	Multi-Layer Perceptron
BCE	Binary Cross-Entropy
HD95	95th-Percentile Hausdorff Distance

mIoU	Mean Intersection over Union
FN	False Negative
FP	False Positive
TN	True Negative
TP	True Positive
AdamW	Adam with Decoupled Weight Decay
CPU	Central Processing Unit
CUDA	Compute Unified Device Architecture
cuDNN	CUDA Deep Neural Network Library
GPU	Graphics Processing Unit
LTS	Long-Term Support
RGB	Red-Green-Blue
CLAHE	Contrast-Limited Adaptive Histogram Equalization
QML	Quantum Machine Learning
QNN	Quantum Neural Network
VQC	Variational Quantum Circuit
PQC	Parameterized Quantum Circuit
Qubit	Quantum Bit
CNOT	Controlled-NOT

General Introduction

Brain tumors are among the most challenging conditions in modern neurology and oncology. Their proliferation within the central organ responsible for regulating virtually every function of the human body makes accurate and timely diagnosis a matter of critical clinical importance [3]. Magnetic resonance imaging (MRI) has become the reference modality for detecting, delineating and monitoring these tumors, owing to its excellent soft-tissue contrast and its absence of ionizing radiation [4]. Yet the interpretation of MRI examinations remains, in current clinical practice, largely dependent on manual analysis by expert radiologists, a process that is time-consuming, subject to inter-observer variability, and increasingly strained by the growing volume of imaging data relative to the availability of specialized clinical expertise [5].

Artificial intelligence, and deep learning in particular, has emerged as a natural response to this bottleneck, enabling models to learn tumor patterns directly from imaging data rather than from handcrafted rules. Convolutional networks, attention mechanisms and Transformer-based architectures have each pushed automated segmentation and classification performance forward, and more recently, large-scale foundation models such as SAM [6] and its medical adaptation, MedSAM [1], have introduced a further shift: rather than training a dedicated network from scratch for every new task, a single pretrained representation, guided by a spatial prompt, can be adapted across a wide range of segmentation problems.

This promptable paradigm is particularly well suited to the medical domain, where annotated data is costly and scarce, but its adaptation to the specific characteristics of brain MRI, and its extension to joint segmentation and classification, remain open and active research questions. At the same time, performance on established segmentation and classification benchmarks has begun to plateau, with successive architectures differing by increasingly narrow margins while requiring an ever-growing number of trainable parameters. This trend motivates the exploration of complementary directions capable of improving, or at least matching, predictive performance without a proportional increase in model complexity. Hybrid quantum-classical machine learning has recently emerged as one such direction: variational quantum circuits exploit superposition and entanglement to represent highly nonlinear transformations using comparatively few trainable parameters [2, 7], raising a concrete question for this work: can a compact quantum component be integrated into a medical foundation-model pipeline without sacrificing the performance that pipeline was built to achieve?

This internship report addresses these two intertwined problems within a single framework. It first develops and rigorously evaluates a MedSAM-based architecture for joint brain tumor segmentation and classification on the BRISC dataset [4], adapting the model through full fine-tuning and a multi-bounding-box prompting strategy designed for tumors of irregular and fragmented shape. It then investigates, within that same framework, whether the classical classification head can be replaced by a variational quantum circuit without degrading classification performance, while substantially reducing the number of trainable parameters involved.

The work was carried out at the PRIME Laboratory of the Université de Moncton, under the Mitacs Globalink program, over a four-month internship structured around three successive experimental phases: the fine-tuning and benchmarking of segmentation architectures, the integration and ablation of the quantum classification head, and the consolidation of both results under a unified, statistically grounded evaluation protocol.

The scope of this work is accordingly twofold: a MedSAM-based framework, benchmarked against retrained baselines and the published state of the art, for the segmentation and classification task on BRISC; and, building on that same framework, a variational quantum classification head evaluated against its classical counterpart under a controlled ablation of the data-embedding strategy, the number of qubits and the circuit depth.

This report is organized into four chapters. Chapter 1 introduces the host institutions, the clinical context of brain tumor analysis, and the problem statement motivating this work. Chapter 2 reviews the relevant literature on deep learning architectures for brain tumor segmentation and classification, situates the BRISC dataset among existing benchmarks, and identifies the limitations that this work aims to address. Chapter 3 presents the proposed MedSAM-based framework in detail, from dataset adaptation and training strategy to its segmentation and classification results. Chapter 4 introduces the necessary quantum computing background, describes the proposed quantum classification head, and reports the ablation studies conducted on the data-embedding strategy, the number of qubits and the circuit depth, before comparing the retained configuration with the classical MLP. The report closes with a general conclusion that synthesizes the contributions of both experimental chapters and outlines directions for future work.

Chapter 1

General Context and Problem Statement

1.1 Introduction

This first chapter situates the internship before entering the technical matter of the report. It begins with the institutional framework in which the work was conducted, namely the Mitacs Globalink program, the Université de Moncton and the PRIME laboratory that hosted the project. It then turns to the medical context: what brain tumors are, why their study matters, how magnetic resonance imaging is used to diagnose them, and what limits the manual interpretation of these images. The objectives of the internship and the schedule followed to reach them close the chapter.

1.2 Host Organization

1.2.1 The Mitacs Program

Mitacs is a Canadian non-profit organization, founded in 1999, aimed at stimulating innovation and strengthening business competitiveness in Canada. By establishing partnerships between universities, companies, and the government, Mitacs facilitates research projects designed to solve concrete real-world problems. The Mitacs Globalink program is particularly distinguished by its commitment to hosting international students, allowing them to collaborate on research projects with local researchers and businesses.

The Mitacs Globalink program represents a true gateway to research in Canada, offering students the opportunity to develop professional skills while expanding their network. This learning experience plays an essential role in preparing students for the challenges of the labor market.

1.2.2 Université de Moncton

Founded in 1963, the Université de Moncton is a French-language university located in New Brunswick, Canada, with campuses in Edmundston, Shippagan, and Moncton, the latter being the largest (Figure 1.1). As the primary French-language post-secondary institution outside of Quebec, the Université de Moncton occupies a central position in Acadian and Francophone society in Canada. It is widely recognized for the quality of its teaching and the diversity of its academic programs, and it hosts several research units active in engineering and applied sciences, among which the laboratory that welcomed this internship.



Figure 1.1: Université de Moncton

1.2.3 PRIME Laboratory

The Perception, Robotics and Intelligent Machines (PRIME) Laboratory was created in the fall of 2017. Since its establishment, PRIME has developed expertise in applied artificial intelligence and conducts research aimed at developing advanced algorithms and innovative solutions in collaboration with industrial partners [8].

The laboratory's research activities cover several application areas, including machine vision, advanced robotics, autonomous navigation, medical imaging and health data, and natural language processing. The present work was conducted within this research environment and is particularly related to PRIME's activities in medical imaging and artificial intelligence.

1.3 Brain Tumor Analysis: Clinical Background and Medical Context

Brain tumors are among the most challenging neurological diseases because they affect the central organ responsible for controlling virtually every function of the human body. They correspond to an abnormal proliferation of cells within the brain or its surrounding tissues and exhibit considerable heterogeneity in terms of biological origin, morphology, growth pattern, clinical behavior, prognosis, and therapeutic response [3]. According to their origin, brain tumors are broadly classified into primary tumors, which develop directly from brain tissues, and secondary tumors (brain metastases), which originate from cancers in other organs before spreading to the brain [9]. Among primary tumors, gliomas represent the most aggressive category and include glioblastoma, the most common malignant primary brain tumor in adults, with a median survival of approximately fifteen months despite multimodal treatment [5]. Other common tumors include meningiomas, which arise from the meninges and are generally benign, and pituitary adenomas, which originate from the pituitary gland and may cause significant endocrine and visual disorders. These three tumor types constitute the focus of this work and correspond to the classes included in the BRISC dataset.

Although brain tumors account for a relatively small proportion of all cancers, they remain one of the deadliest because of their critical location and limited therapeutic options. According to the International Agency for Research on Cancer (IARC), more than 300,000 new brain tumor cases are diagnosed worldwide each year, corresponding to an incidence of approximately 24.8 cases per 100,000 inhabitants. Meningiomas account for nearly 41% of diagnosed tumors, while gliomas represent approximately 26.3% [10]. In Canada alone, approximately 3,300 new cases are expected in 2026, with nearly 50,000 people currently living with a brain tumor and about 2,700 deaths anticipated during the same year [11]. The burden is even greater in developing countries where access to advanced neuroimaging remains limited. A retrospective Tunisian study involving 7,055 cancer patients reported that only 11.5% of patients underwent magnetic resonance imaging (MRI), despite MRI being the reference modality for brain tumor diagnosis, resulting in delayed detection of metastatic lesions and poorer clinical outcomes [12]. These statistics illustrate both the medical importance of brain tumors and the need for efficient diagnostic strategies.

Magnetic Resonance Imaging (MRI) has become the reference imaging modality for the diagnosis and follow-up of brain tumors because it provides high-resolution images of brain anatomy without exposing patients to ionizing radiation [4]. Its excellent soft-tissue contrast enables clinicians to accurately visualize tumor morphology, location, infiltration, edema, and surrounding anatomical structures. A standard clinical examination combines complementary MRI sequences, including T1-weighted, T2-weighted, contrast-enhanced T1 (T1ce), and FLAIR images, each providing specific diagnostic information that supports tumor detection, treatment planning, surgical guidance, radiotherapy, and longitudinal patient monitoring [5]. Consequently, accurate interpretation of brain MRI is fundamental for ensuring appropriate clinical decision-making and improving patient prognosis.

1.4 Problem Statement

Despite the remarkable capabilities of MRI, extracting clinically relevant information from these images remains a challenging and time-consuming task. In current clinical practice, tumor delineation is still largely performed manually by expert radiologists, who must carefully identify tumor boundaries across numerous MRI slices. This process is labor-intensive and inherently subjective, leading to significant inter- and intra-observer variability. Furthermore, brain tumors often exhibit irregular shapes, heterogeneous textures, diffuse infiltration, and poorly defined boundaries, making precise delineation particularly difficult [5]. As the number of MRI examinations continues to increase while the availability of experienced neuroradiologists remains limited, manual analysis is becoming increasingly impractical for routine clinical applications.

These limitations have motivated the development of computer-aided diagnosis systems based on artificial intelligence. Deep learning has already changed how automatic brain tumor segmentation and classification are approached, since models can now learn complex representations directly from imaging data instead of relying on handcrafted rules. Foundation models such as MedSAM and SAM2 have pushed this further, showing that a single pretrained representation can generalize across imaging tasks that would once have required a dedicated network each. Reliable automated systems for brain tumor analysis are nonetheless still far from solved [4]. Their performance depends heavily on large, high-quality annotated datasets, and expert annotation remains expensive and slow to produce at scale. The variability inherent to tumor size, morphology, location, imaging protocol and acquisition device compounds the difficulty, making it hard for a model trained under one set of conditions to generalize to another.

A second, related issue concerns model complexity. State-of-the-art architectures now routinely contain hundreds of millions of parameters, and the computational cost of fine-tuning them has grown accordingly, yet the accuracy gains obtained from one generation of models to the next have become noticeably smaller. Segmentation performance on benchmarks such as BRISC has in fact approached a ceiling close to inter-annotator agreement, leaving comparatively little room for further improvement through architectural refinement alone. Classification, which typically relies on a much smaller component appended to a shared visual representation, is where a

different kind of gain becomes possible: not necessarily higher accuracy, but a substantially smaller and more efficient decision-making component.

Hybrid quantum-classical learning offers one way of pursuing exactly that kind of gain. Variational quantum circuits use superposition and entanglement to represent highly nonlinear transformations with a comparatively small number of trainable parameters, which raises a concrete question for the classification side of a medical imaging pipeline: can a compact quantum circuit reach the same classification performance as a classical head, while requiring only a fraction of its parameters?

This internship investigates that question by developing a MedSAM-based framework for joint brain tumor segmentation and classification on the BRISC dataset. The segmentation branch is addressed through full fine-tuning and a multi-bounding-box prompting strategy adapted to tumors of irregular and fragmented shape, and is benchmarked against a set of classical and Transformer-based architectures retrained under an identical protocol. Building on this shared representation, the classification branch is then used to evaluate whether a variational quantum circuit can replace the classical MLP head without a loss of accuracy, through a controlled ablation of the data-encoding strategy, the qubit count and the circuit depth. The two contributions are assessed independently, and together they determine whether combining a fine-tuned medical foundation model with a lightweight quantum classification head constitutes a viable and resource-efficient direction for brain tumor analysis.

1.5 Project Planning

The internship followed a sequential logic: each phase was designed to produce exactly what the next one would require, so that no later step began before its prerequisites were in place. The opening weeks combined a focused literature review with the setup of the working environment on the laboratory servers, followed by the exploration and preprocessing of the BRISC dataset. This third step was not merely preparatory — it fixed the data partitions and the evaluation protocol shared by every experiment conducted afterwards, since a benchmark is only meaningful once every model it contains is trained and evaluated under identical conditions. Establishing that protocol before any model was trained was accordingly a precondition for the whole comparative study, not an incidental first task.

The benchmark of classical segmentation architectures came next, producing the reference values against which every subsequent model would later be positioned. The study of foundation models followed, centred on the fine-tuning of MedSAM and the investigation of prompting strategies adapted to irregular tumor shapes; this phase in turn produced the specialised representation on which the quantum classification phase depends. That final experimental phase covered the implementation of the variational quantum circuits and their integration into the classification branch, before a dedicated ablation and statistical analysis isolated the contribution of each design choice.

The overlaps visible in Figure 1.2 are deliberate rather than incidental. Since the laboratory server provides four GPUs, several configurations could be trained concurrently, so the final runs of one phase were often still underway while the first experiments of the next were already being prepared. Writing began in the ninth week, once the benchmark results were stable enough to be reported, and continued in parallel with the remaining experiments through to the defence — a choice that avoided concentrating the redaction of the report in the final weeks and left room to revise earlier chapters as later results came in.

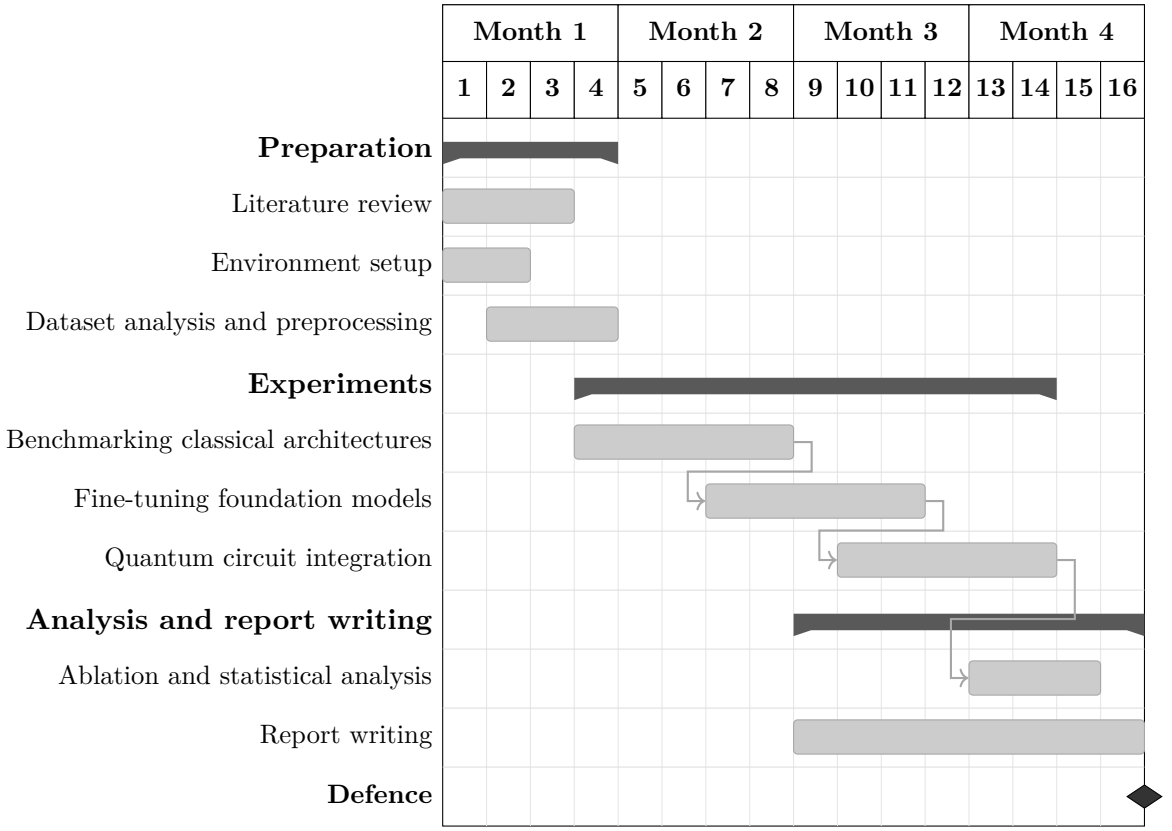


Figure 1.2: Gantt chart showing the project timeline and task dependencies.

1.6 Conclusion

This chapter established the general context and motivation of this work by introducing the medical challenges of brain tumor analysis and the limitations of current automated approaches. The next chapter reviews the state of the art in brain tumor segmentation and classification and presents the BRISC dataset that forms the basis of the experimental study.

Chapter 2

Related Work

2.1 Introduction

Brain tumor analysis from magnetic resonance imaging (MRI) has become a central topic in medical image analysis, driven by the rapid progress of deep learning for automated segmentation and classification. This chapter reviews the approaches reported in the recent literature. It first examines the deep learning architectures proposed for brain tumor segmentation, following their evolution from convolutional networks to attention-based, Transformer-based and foundation models. It then reviews the architectures used for tumor classification and the question of cross-dataset generalization. The metrics used to evaluate both tasks are subsequently formalized, and the main publicly available brain tumor MRI datasets are described and compared. The chapter closes with the limitations identified across the reviewed studies.

2.2 Deep Learning Architectures for Brain Tumor Analysis

Deep learning has progressively transformed automated brain tumor analysis by replacing handcrafted image descriptors with learned hierarchical representations. Two complementary tasks dominate this field: *segmentation*, which aims to delineate the tumor region at the pixel or voxel level, and *classification*, which determines the tumor category from the acquired MRI data. The evolution of deep learning architectures has followed a progressive transition from convolutional neural networks (CNNs), which are particularly effective at learning local spatial features, toward attention mechanisms and Transformer architectures capable of modeling broader contextual dependencies. More recently, foundation models have introduced highly transferable representations that can be adapted to medical imaging through prompting or fine-tuning.

The following sections review representative architectures proposed in recent literature for brain tumor segmentation and classification, with particular attention to their architectural principles, reported results and remaining limitations.

2.2.1 Brain Tumor Segmentation

- **CNN-Based Architectures.**

  Convolutional neural networks remain an important approach for brain tumor image analysis because of their ability to learn hierarchical visual representations from MRI data. A representative segmentation architecture is ZNet, proposed by Ottom *et al.* for automatic two-dimensional brain tumor segmentation [13]. ZNet is inspired by the U-Net encoder–decoder structure and incorporates residual-style feature fusion between the encoder and decoder. Its encoder consists of five blocks, each combining convolutional layers, batch normalization, ReLU activation and max-pooling. A distinctive component of the architecture is the concatenation of encoder input features with the corresponding encoded features after interpolation to matching dimensions. The decoder then progressively reconstructs the segmentation using transposed convolutions, skip connections and double-convolution blocks, followed by a 1×1 convolution and sigmoid activation.

The authors evaluated ZNet on the TCGA-LGG dataset using FLAIR MRI slices. Extensive data augmentation was applied using horizontal and vertical flipping, random rotations, transposition and shift–scale–rotate transformations. On the independent test set, ZNet achieved a Dice score of 0.92, compared with 0.85 for the conventional U-Net baseline [13]. These results indicate that residual feature fusion and extensive augmentation can improve the performance of encoder–decoder CNNs for low-grade glioma segmentation.

CNN architectures have also been investigated for multi-class brain tumor classification. Thahiruddin and Wulandari compared ResNet18, EfficientNet-B0, MobileNetV3-Small and MobileViTV2 on the BRISC dataset [14]. The first three models are CNN-based architectures, whereas MobileViTV2 combines convolutional layers with Transformer blocks. The study considered not only classification accuracy but also computational efficiency and inference speed, with the objective of assessing the suitability of different architectures for practical deployment. The results showed that CNN-based models can provide strong classification performance while maintaining relatively efficient processing.

- **Attention-Enhanced Architectures.**

  Attention mechanisms have been incorporated into convolutional architectures to improve the selection of informative image features. In the BRISC study, Fateh *et al.* evaluated several segmentation architectures, including U-Net, BASNet, SaberNet and ABANet [4]. The investigated models progressively incorporate attention mechanisms to enhance feature representation and improve the localization of tumor regions.

  According to the reported BRISC results, the conventional U-Net baseline achieved a weighted mean Intersection over Union (mIoU) of approximately 75.7%, whereas BASNet, SaberNet and ABANet achieved approximately 79.5%, 80.3–80.4% and 80.6%, respectively [4]. The improvement observed for the attention-enhanced architectures indicates the benefit of emphasizing informative features when segmenting tumor regions against a large background.

  The BRISC study therefore provides evidence that incorporating attention mechanisms into convolutional segmentation architectures can improve performance compared with a conventional U-Net configuration. However, these approaches remain based primarily on convolutional representations, motivating the investigation of architectures capable of modeling broader spatial relationships.

- **Transformer-Based Architectures.**

  Transformer-based architectures have been increasingly investigated for brain tumor analysis because self-attention provides a mechanism for modeling relationships between different image regions. A representative segmentation approach is SegFormer-UAttNet, proposed by Abbas *et al.* for brain tumor segmentation on BRISC [15, 16].

  SegFormer-UAttNet combines a pretrained SegFormer-B5 Vision Transformer encoder with an attention-enhanced U-Net decoder. The SegFormer encoder extracts hierarchical multi-scale representations, while the decoder progressively reconstructs high-resolution tumor

segmentation maps. Feature fusion is performed through skip connections, and Squeeze-and-Excitation modules are incorporated to enhance the decoder's feature representations. Dropout regularization is also employed. The encoder is pretrained on ImageNet and kept frozen during training, reducing the computational requirements and limiting the risk of overfitting [16].

The authors evaluated the architecture on the BRISC segmentation dataset. The preprocessing procedure included resizing the MRI images to 512×512, RGB conversion and normalization using ImageNet statistics. A weighted Tversky loss was used to address the imbalance between tumor and background pixels. SegFormer-UAttNet achieved a Dice coefficient of 0.948, an IoU of 0.903 and a mean AUC of 0.940 [16]. The reported results demonstrate the effectiveness of combining Transformer-based feature extraction with an attention-enhanced encoder–decoder decoder for brain tumor segmentation.

Transformer architectures have also been investigated for brain tumor classification. Thahiruddin and Wulandari included MobileViTV2 in their comparison on BRISC, combining convolutional operations for local feature extraction with Transformer blocks for global contextual representation [14]. In a broader classification study, Viana *et al.* compared ResNet-50, DenseNet-121 and ViT-B16 and investigated their generalization across multiple datasets [17]. Their experiments highlight the importance of evaluating Transformer-based and CNN-based classifiers not only on internal test data but also on external datasets.

- **Hybrid and Cross-Dataset Classification Architectures.**

  Recent classification studies increasingly combine different architectural families rather than relying exclusively on either CNNs or Transformers. Thahiruddin and Wulandari compared conventional CNNs with MobileViTV2, a hybrid CNN–Transformer architecture, on the four-class BRISC classification task [14]. This comparison provides a direct assessment of whether incorporating Transformer-based global representation into a lightweight convolutional architecture can provide an advantage over purely convolutional alternatives.

  Viana *et al.* further investigated ResNet-50, DenseNet-121 and ViT-B16 under different augmentation strategies and evaluated their performance on both BRISC and an external BCD-MRI dataset [17]. This study is particularly relevant because it examines model generalization rather than relying exclusively on performance obtained from the dataset used for training. The authors show that high internal classification performance does not necessarily guarantee comparable performance on an external dataset, emphasizing the importance of cross-dataset validation in medical image classification.

  Taken together, these studies illustrate two complementary trends in recent brain tumor classification research: the continued effectiveness of CNN architectures and the increasing adoption of hybrid or Transformer-based models to capture broader image context. They also highlight that architectural performance should be interpreted together with the evaluation protocol and the ability of a model to generalize beyond its training distribution.

2.2.2 Brain Tumor Classification

While segmentation determines the spatial extent of a tumor, classification aims to determine its pathological category. For MRI-based brain tumor analysis, commonly considered classes include glioma, meningioma and pituitary tumors, with some datasets additionally including non-tumorous cases. Recent research has explored both conventional CNNs and hybrid architectures combining convolutional and Transformer-based representations.

- **CNN-Based Classification Architectures** Convolutional Neural Networks (CNNs) remain among the most widely used architectures for brain tumor classification because of their ability to learn discriminative local and hierarchical image representations. ResNet introduced residual connections that facilitate the optimization of deep networks by providing direct information pathways across multiple layers [18]. DenseNet subsequently introduced dense connectivity between successive layers, encouraging feature reuse and improving gradient propagation, while EfficientNet adopted a compound scaling strategy that jointly adjusts network depth, width, and input resolution to achieve a favorable accuracy–efficiency trade-off.

  Recent experiments on the BRISC dataset demonstrate the effectiveness of these architectures for multi-class brain tumor classification. Fateh *et al.* reported classification accuracies of 99.2% for EfficientNet-B0 and 98.2% for ResNet-50 on BRISC [4]. These results demonstrate that CNN-based models can learn highly discriminative representations from the T1-weighted MRI images provided by the dataset.

  Thahiruddin and Wulandari further investigated several conventional CNN architectures for brain tumor classification on BRISC, comparing their performance with hybrid architectures [14]. Their study illustrates the continued relevance of CNNs for this task while also highlighting the increasing interest in architectures capable of incorporating broader contextual information.

  Despite their strong classification capabilities, CNNs primarily construct feature representations through local receptive fields. Capturing relationships between anatomically distant regions therefore generally requires deeper architectures or additional mechanisms for contextual modeling.

- **Hybrid CNN–Transformer Architectures** The combination of convolutional and Transformer mechanisms has emerged as an important direction for brain tumor classification. CNNs provide effective extraction of local texture and structural patterns, whereas Transformers use self-attention to model relationships between spatially separated image regions. Hybrid architectures consequently aim to combine complementary local and global representations.

  Thahiruddin and Wulandari evaluated hybrid CNN–Transformer architectures alongside conventional CNN models on the BRISC dataset [14]. Their comparison reflects the growing adoption of Transformer-based representations for brain tumor classification and the interest in combining convolutional feature extraction with attention-based contextual modeling.

More recent research has also emphasized generalization and robustness rather than relying exclusively on performance obtained from an internal test set. Viana *et al.* investigated deep learning-based brain tumor classification using external validation across multiple datasets [17]. Their study emphasizes that models achieving high accuracy on a single dataset may experience performance degradation when applied to images acquired from different populations, institutions, or imaging protocols. External validation is therefore an important consideration when assessing the robustness and potential clinical applicability of brain tumor classification models.

- **General Trends in Brain Tumor Classification** The evolution of brain tumor classification architectures follows a progression from conventional CNNs toward hybrid and Transformer-based approaches. CNNs remain attractive because of their efficient local feature extraction and established performance, whereas Transformer-based architectures provide mechanisms for modeling broader contextual relationships. Hybrid models attempt to combine these complementary properties within a unified architecture [14].

  At the same time, recent studies increasingly emphasize generalization, robustness, and computational efficiency in addition to classification accuracy. The results reported on BRISC demonstrate that modern CNN architectures can already achieve very high classification performance [4], while studies involving external validation underline the importance of evaluating models beyond a single curated dataset [17]. Consequently, the literature does not indicate a universally optimal architecture; rather, the choice of model involves a trade-off between discriminative capability, contextual modeling, computational cost, and generalization to heterogeneous MRI data.

2.3 Brain Tumor MRI Datasets

Deep learning models for brain tumor analysis depend on the availability of large, diverse and accurately annotated datasets, which also provide the standardized benchmarks on which competing methods can be compared. Several such resources have been released over the past decade. They differ in the tumor types they cover, in the MRI modalities they provide and, most importantly for the present work, in whether they support segmentation, classification, or both.

A first family of datasets is dedicated to glioma segmentation. The Brain Tumor Segmentation (BraTS) benchmark, introduced in 2012 and updated through annual challenges, provides multimodal examinations from several institutions with expert annotations of the enhancing tumor, the tumor core and the whole tumor [19]. Its four complementary modalities, its standardized preprocessing and its international challenge have made it the reference for volumetric segmentation. That same preprocessing, however, involving skull stripping, registration and intensity normalization, removes much of the variability encountered in routine practice, and the restriction to gliomas excludes any multi-class formulation. The Medical Segmentation Decathlon reuses the same material in its brain tumor task [20], and the Cancer Genome Atlas Low-Grade Glioma collection (TCGA-LGG) offers FLAIR examinations with expert masks, linked to genomic and clinical records, which makes it particularly suited to radiogenomic studies [21]; several convolutional architectures have reported competitive segmentation performance

on it [13]. Both share the limitation of the family: a single tumor type. The Cancer Imaging Archive, for its part, is not a dataset but an aggregator of heterogeneous collections [22], so it provides no unified annotation protocol.

A second family addresses classification rather than delineation. The dataset introduced by Cheng *et al.* covers three categories of primary brain tumor, glioma, meningioma and pituitary adenoma, and has been widely used to evaluate multi-class convolutional and Transformer models [23]. It broadens the scope of the task, but its limited size, its class imbalance and its homogeneous acquisition conditions restrict generalization, and segmentation masks are not available for all of its images [4]. Segmentation and classification therefore remain separated: the first family annotates boundaries without offering classes, the second offers classes without complete boundaries.

The Brain Tumor Segmentation and Classification (BRISC) dataset was introduced to close this gap by supporting both tasks within a single benchmark. It gathers approximately 6,000 contrast-enhanced T1-weighted images from several public repositories, among them Figshare, SARTAJ and Br35H, organized into four balanced categories, and its segmentation subset provides 4,793 images with pixel-wise binary masks validated through a multi-stage radiological review, for an inter-annotator agreement of 0.924 in Dice [4]. The scans come from three anatomical planes and from different institutions and devices, which introduces substantial variability in tumor size, location and appearance. Its acknowledged limitations are the restriction to a single modality, the absence of detailed acquisition metadata and a possible domain shift between contributing institutions.

Table 2.1 summarizes these resources. The last two columns carry the argument: among the datasets reviewed, only BRISC combines a large number of images, balanced classes, expert pixel-level masks and classification labels for several tumor types, which explains its rapid adoption as a benchmark for both tasks [15, 14, 17] and motivates its selection for this work.

Table 2.1: Comparison of publicly available brain tumor MRI datasets.

Dataset	**Year**	**Tumor types**	**Modalities**	**Masks**	**Class labels**
BraTS [19]	2012–2025	Gliomas (HGG/LGG)	T1, T1ce, T2, FLAIR	Yes	No
MSD Task01 [20]	2018	Gliomas	T1, T1ce, T2, FLAIR	Yes	No
Cheng [23]	2015	Glioma, meningioma, pituitary	T1ce	Partial	Yes
TCGA-LGG [21]	2019	Low-grade gliomas	FLAIR	Yes	No
TCIA [22]	2011–present	Multiple collections	Multi-modal	Variable	Variable
BRISC [4]	2025/2026	Glioma, meningioma, pituitary, no tumor	T1ce	Yes	Yes

2.4 Limitations of Existing Approaches

Two general observations emerge from the studies reviewed above. For segmentation, the transition from convolutional to attention-based and then Transformer-based representations produces a consistent improvement, the best published result on BRISC being obtained by SegFormer-UAttNet with a Dice coefficient of 0.948 and an IoU of 0.903 [15]. For classification, by contrast, the reported accuracies are concentrated within a narrow band between 96.9% and 99.3%, and the differences among the leading models are frequently reported as statistically insignificant [14], which suggests that the discriminative capacity of the task is close to being saturated on this benchmark.

Beyond these observations, the reviewed literature reveals several recurring limitations. First, segmentation and classification are generally addressed as independent problems, with distinct models trained separately, which increases the overall computational cost and prevents the two tasks from sharing a common representation. Second, most published results are obtained under heterogeneous experimental protocols, with different data splits, augmentation strategies and training schedules, which makes reported performances difficult to compare rigorously; the coexistence of mIoU, Dice and IoU as headline metrics across studies further complicates direct comparison. Third, performance on established benchmarks appears to be approaching saturation, successive architectures differing by increasingly small margins obtained at the cost of a substantial increase in the number of trainable parameters. Fourth, external validation remains rare, whereas the results of Viana *et al.* show that a model reaching more than 99% accuracy internally may fall below 89% on an external dataset [17]. Finally, foundation models, despite their strong transfer capabilities, remain insufficiently evaluated on brain tumor MRI, and the interactive nature of prompt-based segmentation is rarely taken into account when comparing them with fully automatic architectures.

2.5 Conclusion

This chapter traced the evolution of brain tumor analysis from convolutional networks to attention-based, Transformer-based and foundation models, described the metrics used to evaluate segmentation and classification, and reviewed the datasets available for these tasks. It also identified the main limitations of existing work, in particular the separation of the segmentation and classification tasks, the heterogeneity of evaluation protocols, the scarcity of external validation and the saturation of performance on established benchmarks. The next chapter builds on this analysis to present the methodology adopted in this study, and compares the results it obtains with the architectures reviewed here.

Chapter 3

Proposed MedSAM-Based Methodology

3.1 Introduction

Medical image analysis requires models that can capture complex anatomical structures while adapting to the characteristics of clinical data. Foundation models such as MedSAM provide a promising basis for addressing these challenges through their pretrained medical representations and prompt-driven segmentation capabilities. This chapter presents the proposed MedSAM-based framework for brain tumor analysis on the BRISC dataset. It describes the adopted architecture, its adaptation to the dataset, the proposed segmentation and classification pipeline, the training strategy, and the experimental evaluation.

3.2 Theoretical Background: the MedSAM Model

This section introduces MedSAM, the segmentation model underlying the proposed framework. It first presents the transition from SAM to MedSAM, then describes its main architectural components and finally explains the prediction mechanism used to generate segmentation masks from an image and a spatial prompt.

3.2.1 From SAM to MedSAM

The Segment Anything Model (SAM) introduced a *promptable* formulation of segmentation, in which the region of interest is not inferred implicitly by the network but designated explicitly through a spatial prompt such as a point or a bounding box [6]. Having been trained on more than one billion masks, the model provides a general-purpose visual representation that transfers to a wide range of segmentation problems without requiring a dedicated architecture to be designed for each of them. The formulation also changes the nature of the learning problem: the network is not asked which object should be segmented, only where its boundary lies.

Transposing this behaviour to medical imaging is not immediate. Evaluations of SAM on clinical data report a degradation on targets characterised by weak boundaries or low contrast, precisely the regime in which most anatomical structures fall. MedSAM was introduced to close this domain gap by fine-tuning SAM on a curated collection of 1,570,263 medical image–mask pairs spanning ten imaging modalities and more than thirty cancer types [1]. What changes is the content of the representation, re-estimated on medical data, and not the way the model is queried.

Two properties of this design matter for the present work. The model remains promptable rather than fully automatic, so the operator retains explicit control over the target region. And among the admissible prompts, the bounding box is privileged, because a rectangle conveys the extent of the lesion and not only a point inside it. Both properties are inherited without modification by the framework proposed in this chapter.

3.2.2 Architectural Components

MedSAM keeps the three-component decomposition of SAM: an image encoder, a prompt encoder and a mask decoder. The decomposition is functional rather than cosmetic, since it isolates the extraction of visual features from the prompt-guided production of a mask, and

the three modules are neither of the same size nor trained in the same way. No convolutional encoder–decoder of the U-Net family intervenes anywhere in this design.

- **Image encoder.** The encoder is a ViT-B backbone [24], and it is by far the largest component of the model. The input is standardised to $1024 \times 1024 \times 3$; a grayscale acquisition such as an MRI slice, which is natively single-channel, is therefore replicated three times to match this format. The image is then divided into patches of 16×16 pixels, which yields a grid of 64×64, that is, 4,096 tokens, each of them obtained by a linear projection of its patch. This sequence traverses the twelve Transformer blocks of the ViT-Base configuration, each block combining multi-head self-attention, a feed-forward layer and layer normalisation, so that every token is progressively enriched with information coming from the rest of the image. The output is a dense feature map of 256 channels over the 64×64 grid, written $256 \times 64 \times 64$: one descriptor of dimension 256 for every 16×16 region of the original slice. This representation, produced by an encoder of 89,670,912 parameters, is the object on which everything else in the model, and everything proposed in this chapter, is built.

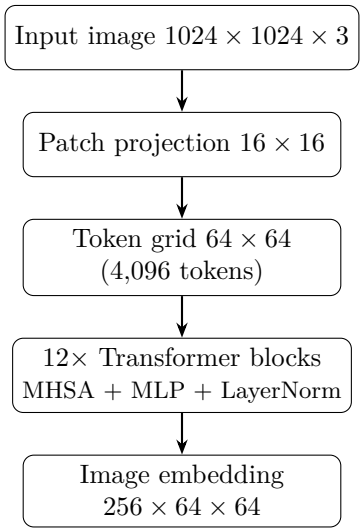


Figure 3.1: Structure of the ViT-B image encoder of MedSAM.

- **Prompt encoder.** This module translates the spatial instruction into a form the decoder can attend to. A bounding box is given by the coordinates of its two opposite corners, top-left and bottom-right, and each corner is converted by positional encoding into an embedding of dimension 256, the same dimension as the image descriptors. The distinction to be underlined is that the prompt encoder never looks at the image content enclosed by the box: it receives four coordinates and returns two vectors, so it states where to look without saying anything about what is there. This is also the only component that the original training procedure leaves untouched [1], its function being entirely determined by the geometry of the box.

- **Mask decoder.** The decoder is the module that confronts the two representations, and it remains deliberately lightweight, with 4,058,340 parameters, roughly twenty times

fewer than the encoder. It receives the image embedding, the prompt embeddings and an additional learned output token, and passes them through two Transformer layers in which cross-attention lets the prompt attend to the image and the image attend to the prompt, so that the visual features situated inside the designated region are progressively selected. Two transposed convolution layers then raise the spatial resolution of the fused representation to 256×256, a sigmoid converts the result into probabilities, and a bilinear interpolation brings the map back to 1024×1024, so that the predicted mask has exactly the size of the input image.

3.2.3 Prediction Mechanism

For a given slice, a prediction is obtained through the following sequence:

1. the image is normalised and resized to $1024 \times 1024 \times 3$;
2. the ViT-B encoder produces the dense embedding;
3. the prompt encoder converts the bounding box into prompt embeddings;
4. the mask decoder fuses the two representations;
5. the predicted mask is returned at the resolution of the input image.

The interesting property of this sequence lies in the way the cost is distributed over it. Almost the entirety of the computation is concentrated in step 2, which mobilises close to 90 million parameters, whereas steps 3 to 5 together mobilise about 4 million: decoding a mask is therefore an operation of a completely different order of magnitude from encoding an image. The second half of the property is that the embedding produced at step 2 is independent of the prompt supplied at step 3, since the encoder never sees the box. The two observations combine into a practical consequence: one image can be encoded once and then decoded several times, under several distinct prompts, for an additional cost that remains negligible with respect to the initial encoding pass. This is a structural property of the architecture rather than an implementation trick, and the prompting strategy introduced in Section 3.3.4 rests entirely upon it. “‘latex

3.3 Proposed Framework and Data Processing Pipeline

This section presents the complete pipeline developed in this work, from the adaptation of the BRISC dataset to the final segmentation and classification outputs. The proposed framework is built around a shared image representation produced by MedSAM and subsequently exploited by two task-specific branches. The following subsections describe the overall architecture, dataset adaptation, preprocessing and augmentation strategy, multi-bounding-box prompting mechanism, and classification branch.

3.3.1 Overview of the Proposed Framework

The framework proposed in this work organises the model around a single shared representation feeding two outputs. An MRI slice is preprocessed and passed to the image encoder, which

produces the dense embedding described in Section 3.2.2. This embedding constitutes the pivot of the architecture, and two branches are derived from it.

The segmentation branch transmits the embedding to the mask decoder, together with the encoded bounding-box prompts, and returns a binary tumor mask. The classification branch reduces the same embedding by global average pooling into a 256-dimensional descriptor and maps it, through a lightweight head, onto the four BRISC categories. Since both branches read from the same tensor, the representation used to delineate the lesion boundary is exactly the representation used to determine the tumor type, and no second backbone is instantiated. The configuration most frequently encountered in the literature reviewed in Chapter 2 proceeds differently, delegating the two tasks to two independently trained networks.

Three modifications separate this framework from the original model. The prompting protocol is replaced by a decomposition of the target into several bounding boxes (Section 3.3.4); the adaptation is conducted as a full fine-tuning in two stages rather than as a decoder-only transfer (Section 3.4.1); and a classification head is grafted onto the shared embedding (Section 3.3.5). Figure 3.2 gives the resulting architecture.

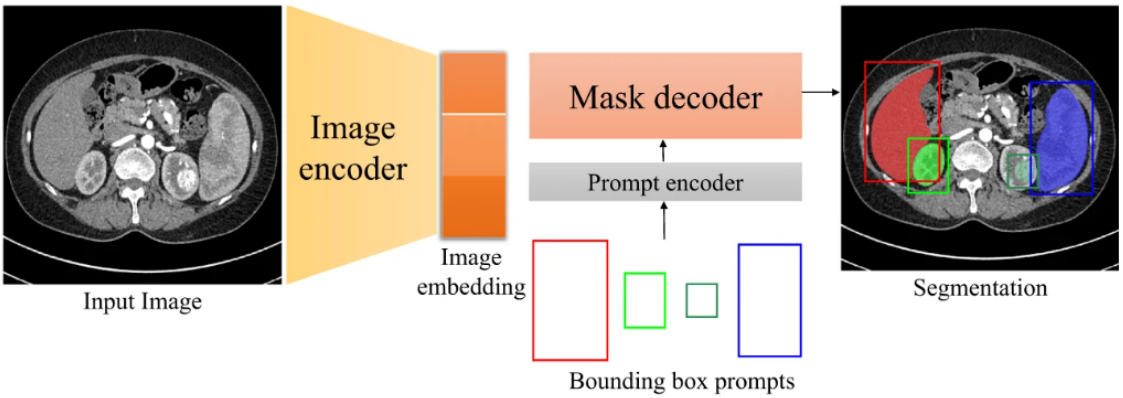


Figure 3.2: Architecture of the proposed MedSAM-based framework [1].

3.3.2 Adaptation of the BRISC Dataset

BRISC is distributed as two independent tasks, and a joint pipeline cannot be trained before they are reconciled. The segmentation subset provides 4,793 tumor images with their binary masks, partitioned into 3,933 training and 860 testing samples; images are stored as `.jpg` files and masks as `.png` files sharing an identical base filename, which allows every image to be paired automatically with its annotation. The classification subset provides 6,000 images distributed over four categories, one of which, *no tumor*, has no counterpart in the segmentation subset.

The unification proceeds as follows. For the images of the segmentation subset, the diagnostic label is recovered directly from the filename, which encodes the tumor type; these samples therefore possess both a mask and a label and populate the glioma, meningioma and pituitary categories. They are joined by the 1,067 *no tumor* training images of the classification subset, which possess a label but no mask. The combined pool amounts to 5,000 images, split at random

under a fixed seed into 4,250 training and 750 validation samples. The predefined test partitions are left untouched and supply 860 images for the evaluation of segmentation and 1,000 images for the evaluation of classification.

The heterogeneity of the annotations is handled by a per-sample binary indicator $\nVdash_{\text{mask}}$. When a sample carries no mask, the forward pass stops after the encoder and the pooling operation: no prompt is constructed, the mask decoder is never invoked, and only the classification term contributes to the loss. This mechanism is not a convenience but a necessity, since interpreting a missing annotation as an empty ground-truth mask would teach the model that a substantial fraction of the brains contains nothing to segment, and would install a systematic bias towards under-segmentation.

3.3.3 Preprocessing and Data Augmentation

The preprocessing pipeline was selected experimentally, by comparing candidate configurations on the validation split. A configuration associating skull stripping with contrast-limited adaptive histogram equalisation (CLAHE) was evaluated and set aside: it did not improve the validation curves, and it introduced an additional failure mode whenever the skull-stripping step behaved incorrectly on sagittal or coronal acquisitions.

The retained pipeline is deliberately minimal and reproduces the convention under which the pretrained weights were optimised [1]. Every image is converted to three channels to match the expected input format, resized to 1024×1024 by bilinear interpolation, and normalised to the range $[0, 1]$ by a per-image min–max transformation; masks are resized by nearest-neighbour interpolation, so that their binary nature survives the operation, and re-binarised afterwards at half the intensity range. Matching the original normalisation is critical rather than incidental. Applying at inference a normalisation different from the one used during fine-tuning displaces the input distribution away from the regime for which the encoder weights were estimated, and the Dice coefficient collapses accordingly.

Augmentation is applied online during training and combines three transformations: a horizontal flip with probability 0.5, a random rotation within $\pm 15^\circ$ with probability 0.5, and a multiplicative brightness perturbation whose factor is drawn uniformly in $[0.85, 1.15]$ with probability 0.4. Geometric transformations are applied jointly to the image and to its mask. One detail governs the interaction between augmentation and prompting: the bounding boxes are extracted *after* augmentation, from the transformed mask, so that the prompt remains geometrically consistent with the image it describes and varies from one epoch to the next for the same sample. The decoder is thus confronted with a continually renewed prompt geometry, which reproduces the intent of the random box perturbation employed in the original training protocol [1]. Images of the *no tumor* class, carrying no mask, receive only the photometric transformation and the flip.

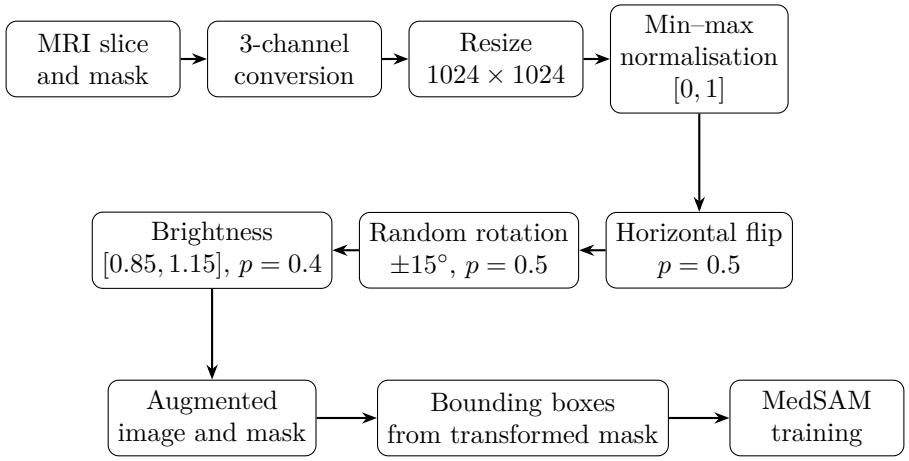


Figure 3.3: Preprocessing, online data augmentation, and prompt generation pipeline.

3.3.4 Multi-Bounding-Box Prompting

In the original protocol, a single bounding box is drawn around the whole target, simulated from the expert annotation [1]. This convention works well when the tumor forms one compact blob, but it becomes imprecise when the tumor is split into several separate pieces or when its shape is strongly irregular. A single rectangle enclosing everything also encloses a large amount of healthy tissue, so the decoder receives a prompt that no longer points at the lesion but at the region that happens to contain it, and the spatial guidance is weakened precisely in the cases where it matters most.

The proposed pipeline therefore builds one box per piece of the tumor instead of one box for the whole tumor. The ground-truth mask is first split into its separate regions, using 8-connectivity so that two tumor pixels touching by a side or by a corner belong to the same region. Regions smaller than 50 pixels are discarded. Each remaining region then receives its own rectangle, which is simply the smallest rectangle containing all its pixels, enlarged by a margin of 10 pixels on each side:

$$b = (x_{\min} - 10,\ y_{\min} - 10,\ x_{\max} + 10,\ y_{\max} + 10). \tag{3.3.1}$$

Here $x_{\min}$ and $x_{\max}$ are the positions of the leftmost and rightmost pixels of the region, and $y_{\min}$ and $y_{\max}$ those of its topmost and bottommost pixels. Together they give the two corners of the rectangle, which is then clipped so that it never leaves the image. Repeating this for every retained region gives the set of boxes $\mathcal{B} = \{b_1, \ldots, b_K\}$, where K is the number of pieces of the tumor. The complete procedure is summarised in Algorithm 3.1.

Table 3.1: Multi-bounding-box prompt generation.

Input: binary mask M; margin $m = 10$ pixels; minimum area $a_{\min} = 50$ pixels
1. Extract the connected components of M with 8-connectivity.
2. Discard every component smaller than $a_{\min}$.
3. For each remaining component, take the smallest rectangle containing it, expand it by m pixels on each side, and clip it to the image bounds.
4. **Output:** the set of boxes $\mathcal{B} = \{b_1, \ldots, b_K\}$.

The difference between the two conventions is visible directly on the images. Figure 3.4 shows the same MRI slice, whose tumor is made of several separate pieces, prompted in the two ways. In the upper image, the original protocol produces one rectangle that has to reach from the leftmost pixel to the rightmost pixel of the whole tumor, and from its topmost to its bottommost pixel. That rectangle therefore also covers everything lying between the pieces, which is healthy tissue: the prompt designates a wide area of the brain rather than the lesion itself, and the decoder is left to decide alone which parts of that area are tumor. In the lower image, the proposed strategy gives each piece its own rectangle. Every box now surrounds one piece closely, the space separating the pieces is no longer designated, and the surface claimed by the prompt is much closer to the true surface of the tumor. The information transmitted to the decoder is identical in nature, a set of rectangles, but considerably more precise in content.



(a) Single bounding box.



(b) Multiple bounding boxes.

Figure 3.4: Comparison of single- and multi-bounding-box prompting for a fragmented tumor.

The two parameters are chosen for complementary reasons. The margin prevents the prompt from coinciding with the answer: without it, the rectangle would touch the lesion exactly, every pixel lying just outside the annotation would also fall outside the prompt, and the model would have no room to place the boundary. Ten pixels add a thin band of surrounding tissue, so that the boundary is decided inside the box rather than on its edge; at the working resolution this represents about one percent of the image width, enough to detach the box from the lesion and too little to bring back the healthy tissue that the strategy is meant to exclude. The minimum area, for its part, removes the isolated specks of a few pixels that manual annotation leaves behind. Such a speck is not a tumor region, yet it would generate a prompt of its own, and since the predictions are combined by a maximum, a spurious region could only be added to the final mask and never removed. A threshold of fifty pixels discards these artefacts while remaining well below the size of a genuine tumor region. Both values are fixed once and applied identically to every image.

Each box is encoded separately and sent to the mask decoder together with the shared image embedding, which returns one probability map per box after interpolation to the input resolution. These maps are then merged pixel by pixel, each pixel keeping the highest probability it received

among all the boxes:

$$p(x) = \max_{k \in \{1,\dots,K\}} p_k(x), \tag{3.3.2}$$

where $p_k(x)$ is the probability given to the pixel x by the box b_k, and $p(x)$ the final probability retained for that pixel. This behaves as a union: a pixel is declared tumoral as soon as at least one box claims it, so no piece of the tumor can be erased by the predictions of the others. The merged map is then clamped away from 0 and 1 and converted back into logits, so that the segmentation loss is computed on a numerically stable quantity.

Two properties follow from this design. The amount of spatial guidance adapts itself to the shape of the lesion, since a compact tumor has a single region and therefore a single box, in which case the strategy reduces exactly to the original convention, while a fragmented tumor receives one box per piece; the method can thus never do worse than the protocol it replaces. The additional cost is also negligible, because the expensive encoding pass is performed once and shared by all the boxes, only the lightweight decoder being re-evaluated for each of them, in direct application of the property established in Section 3.2.3.

A safeguard covers the rare case in which a mask produces no valid box after filtering. A small default box is then placed at the centre of the image and the sample is logged, so that these cases can be inspected afterwards instead of silently corrupting the training signal.

3.3.5 Classification Branch

The dense embedding produced by the encoder is reduced by global average pooling into a 256-dimensional descriptor, which is submitted to the classification head and mapped onto the four BRISC categories: glioma, meningioma, pituitary tumor and no tumor. Three consequences follow from this placement. The branch consumes the embedding and not the predicted mask, so a segmentation failure does not propagate mechanically into the class decision. It requires no prompt and invokes neither the prompt encoder nor the mask decoder, which means that classification operates in a fully automatic regime although segmentation remains prompt-driven. And because no box is needed, the *no tumor* images traverse exactly the same code path as the others, which is what makes the joint formulation of Section 3.4.2 possible at all.

The head itself, illustrated in Figure 3.5, is a lightweight multi-layer perceptron: a layer normalisation applied to the pooled descriptor, a fully connected layer projecting 256 units onto 128 followed by a GELU activation and a dropout of probability 0.1, and a final fully connected layer emitting the four class logits. It totals 33,924 trainable parameters, less than 0.04% of the complete model. This deliberate simplicity keeps the architectural complexity where it belongs, inside the segmentation framework, and it serves a second purpose that becomes decisive later: the head being the only module standing between a fixed representation and the class decision, it constitutes a controlled substrate in which an alternative head can be substituted and evaluated in isolation, which is the experiment conducted in Chapter 4.

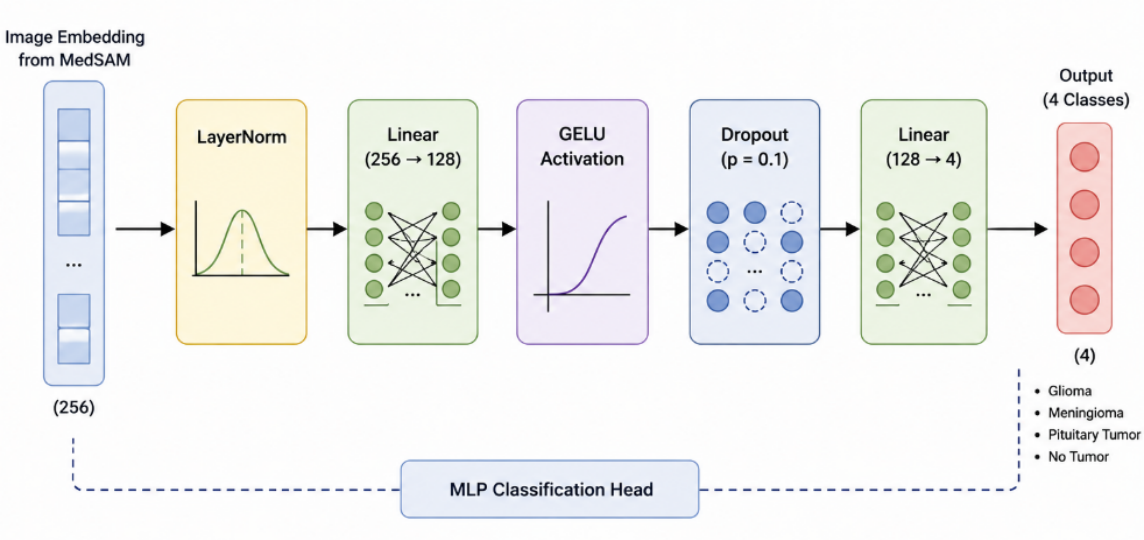


Figure 3.5: MLP head for classification.

3.4 Training Strategy

3.4.1 Two-Stage Full Fine-Tuning

Most adaptations of SAM-family models to a downstream domain freeze the image encoder and train the mask decoder alone, a choice justified by the size of the encoder and by the fear of degrading a representation acquired at considerable cost. The configuration is nevertheless limited by construction: freezing the encoder means that the description of brain MRI remains the generic medical description learned during pretraining, and that only its decoding is specialised. The authors of MedSAM themselves adopted the opposite position when transferring SAM to the medical domain, freezing only the prompt encoder, whose function is already fully determined by the box, and updating the parameters of both the image encoder and the mask decoder [1].

The same principle is applied here, in two stages.

Stage 1 — full fine-tuning on BRISC. The image encoder and the mask decoder are jointly optimised on the segmentation subset with multi-bounding-box prompts. Updating some 90 million parameters on a few thousand images requires the optimisation to be constrained, which is obtained by assigning the pretrained encoder a learning rate on the order of 10^{-6}, one to two orders of magnitude below the value used in the original training procedure [1]. Under such a step size the representation is refined rather than overwritten. This stage produces the specialised checkpoint on which the second stage operates.

Stage 2 — joint training of the two branches. The full set of weights obtained at the end of Stage 1 is reloaded and the image encoder is frozen. The representation is by then already specialised to brain MRI, and it is held fixed while the prompt encoder, the mask decoder and the classification head are trained together under the objective defined below. Freezing at this point has a material justification alongside the methodological one, since the encoder operating at 1024×1024 already saturates the memory of the device used, and keeping its parameters

and its optimiser state out of the computational graph is what renders joint training feasible on the available hardware. It also fixes the representation presented to the classification head, a property exploited in Chapter 4 to isolate the effect of the head alone.

3.4.2 Joint Objective

The two branches are optimized within a single training loop rather than in two successive phases. Every sample contributes to the classification objective, while samples with valid segmentation masks additionally contribute to the segmentation objective. The total training loss is defined as

$$\mathcal{L}_{\text{total}} = \mathbb{1}_{\text{mask}}\, \mathcal{L}_{\text{seg}} + \lambda\, \mathcal{L}_{\text{cls}}, \tag{3.4.1}$$

where $\mathbb{1}_{\text{mask}}$ is a per-sample indicator specifying whether a valid segmentation mask is available, and $\lambda = 0.5$ controls the contribution of the classification loss. This formulation allows the shared image representation to be optimized simultaneously for pixel-level segmentation and image-level tumor classification.

The two components of the objective are defined as follows:

- **Segmentation loss.**

  The segmentation objective combines Dice loss and binary cross-entropy with equal weights:

  $$\mathcal{L}_{\text{seg}} = 0.5\, \mathcal{L}_{\text{Dice}} + 0.5\, \mathcal{L}_{\text{BCE}}. \tag{3.4.2}$$

  For a predicted probability map $\hat{y}$ and a ground-truth mask y defined over N pixels, the soft Dice loss is given by

  $$\mathcal{L}_{\text{Dice}} = 1 - \frac{2\sum_{i=1}^{N} \hat{y}_i y_i + \varepsilon}{\sum_{i=1}^{N} \hat{y}_i + \sum_{i=1}^{N} y_i + \varepsilon}, \tag{3.4.3}$$

  while the binary cross-entropy is defined as

  $$\mathcal{L}_{\text{BCE}} = -\frac{1}{N} \sum_{i=1}^{N} \left[y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i) \right]. \tag{3.4.4}$$

  The combination of the two terms provides complementary supervision. The Dice component directly promotes overlap between the predicted and reference regions, which is particularly relevant for the relatively small tumor regions in MRI slices. The BCE component provides pixel-level supervision and remains informative when the predicted foreground region is small. The smoothing term ε improves numerical stability, particularly when both the prediction and the reference contain little or no foreground.

- **Classification loss.**

  The classification branch uses the standard multi-class cross-entropy:

$$\mathcal{L}_{\text{cls}} = -\sum_{c=1}^{C} y_c \log(\hat{y}_c), \tag{3.4.5}$$

where $C = 4$ corresponds to the four BRISC categories: glioma, meningioma, pituitary tumor, and no tumor. Here, y_c denotes the one-hot target and $\hat{y}_c$ represents the predicted probability for class c. No class weighting is applied, as the resulting training pool is sufficiently balanced across the four categories.

3.4.3 Optimisation

Optimisation relies on the AdamW algorithm with a weight decay of 10^{-4} and distinct parameter groups, so that the segmentation modules and the classification head can be assigned separate learning rates. The prompt encoder and the mask decoder are optimised at 5×10^{-6} and the classification head at 10^{-4}; the image encoder, frozen at this stage, is excluded from the optimiser altogether. Each group follows a cosine annealing schedule decaying smoothly towards zero over the course of training, an arrangement that lets the randomly initialised head take large steps early on while the pretrained components converge with steps small enough to preserve what they already encode.

Training is conducted under automatic mixed precision with a dynamic loss scaler. At an input resolution of 1024×1024 a single sample requires close to 11 GB of GPU memory, so mixed precision is not an optimisation but a condition of feasibility. Images are consequently processed one at a time, gradients being accumulated over four consecutive steps before an optimiser update, which yields an effective batch size of four. Gradient clipping bounds the norm of the accumulated update to 5.0 and prevents the instabilities that the large early gradients of an untrained head would otherwise induce.

At the end of every epoch the model is evaluated on the validation split and a checkpoint is retained whenever the selection criterion improves. Early stopping interrupts training after ten consecutive epochs without improvement, and all the results reported below are obtained from the best checkpoint rather than the last one.

3.4.4 Training Configuration

The training configuration was selected to provide a stable adaptation of MedSAM to the BRISC dataset while taking into account the computational requirements of processing MRI images at high resolution. The main configuration is summarized in Table 3.2.

The input resolution was set to 1024×1024, following the resolution used by MedSAM and allowing the model to preserve fine tumor boundaries that could be lost at lower resolutions. The original SAM ViT-B backbone was retained and fully fine-tuned during the segmentation stage so that its pretrained medical representation could be adapted to the characteristics of BRISC. The proposed multi-bounding-box strategy was used to provide more localized spatial

guidance for tumor regions containing multiple connected components.

AdamW was selected as the optimizer because it provides decoupled weight decay and stable optimization for Transformer-based architectures. A relatively small learning rate of 5×10^{-6} was used for the pretrained segmentation components to avoid excessively modifying the learned representation during fine-tuning. In contrast, the classification head was initialized for the BRISC task and therefore used a higher learning rate of 10^{-4} to allow it to adapt more rapidly.

Training was limited to a maximum of 50 epochs, with early stopping after 10 consecutive epochs without improvement on the validation criterion. This prevents unnecessary training once the model has converged and reduces the risk of overfitting. Because of the memory requirements of the 1024×1024 input resolution, images were processed with an effective batch size of 4 using gradient accumulation.

The segmentation objective combines Dice loss and binary cross-entropy with equal weights. This choice provides both region-overlap supervision and pixel-level supervision, which is suitable for the highly imbalanced foreground-background nature of tumor segmentation. The classification loss is weighted by $\lambda = 0.5$ so that it contributes to the joint objective without dominating the segmentation task, which remains the primary objective of the framework.

The training and validation sets contain 4,250 and 750 images, respectively. This split provides sufficient samples for model optimization while retaining a separate validation subset for model selection and early stopping. The test sets are kept completely independent from training and validation, with 860 images used for segmentation evaluation and 1,000 for classification evaluation.

Table 3.2: Main training configuration of the proposed MedSAM-based framework.

Parameter	**Value**
Backbone	MedSAM (SAM ViT-B), full fine-tuning
Input resolution	$1024 \times 1024 \times 3$
Prompt	Multiple bounding boxes
Optimizer	AdamW
Learning rate	5×10^{-6} (segmentation), 10^{-4} (classification head)
Epochs	50 maximum, early stopping (patience = 10)
Effective batch size	4
Segmentation loss	$0.5\times$ Dice $+ 0.5\times$ BCE
Classification loss	Cross-entropy, weight $\lambda = 0.5$
Training / validation split	4,250 / 750
Test set	860 (segmentation), 1,000 (classification)

3.4.5 Computational Environment

The experiments were carried out on *obiwan*, a high-performance computing server of the PRIME laboratory, accessed remotely over SSH. The machine runs Ubuntu 24.04.4 LTS on an AMD Ryzen Threadripper PRO 5995WX processor offering 64 physical cores and 128 threads, with approximately 500 GiB of system memory. Acceleration is provided by four NVIDIA GeForce RTX 4090 cards of 24 GB each, which allows several configurations to be trained concurrently, one per device.

A single characteristic of the model dictated the shape of the training protocol. MedSAM ViT-B operates at the fixed resolution imposed by its positional embeddings, and the memory footprint of one sample at that resolution rules out conventional mini-batching, hence the single-image batches combined with gradient accumulation described in Section 3.4.3. The software stack was isolated in a dedicated Python virtual environment, with PyTorch 2.5.1 and CUDA 12.1 as the deep learning framework. Table 3.3 details the hardware and software specifications.

Table 3.3: Computational environment used for the experiments.

Component	**Specification**
Hardware	
Computational server	obiwan (PRIME laboratory)
Operating system	Ubuntu 24.04.4 LTS
CPU	AMD Ryzen Threadripper PRO 5995WX
CPU cores / threads	64/128
System memory	499 GiB
GPUs	4× NVIDIA GeForce RTX 4090
GPU memory	24,564 MiB per GPU
NVIDIA driver	570.148.08
Software	
Python	3.12.3
PyTorch	2.5.1+cu121
TorchVision	0.20.1+cu121
CUDA / cuDNN	12.1/9.1.0
PennyLane	0.45.0
NumPy	2.5.0
OpenCV	4.13.0
scikit-learn	1.9.0
scikit-image	0.26.0

Experiments were launched as detached background processes so as to survive an SSH disconnection, each one pinned to a specific device through the `CUDA_VISIBLE_DEVICES` environment variable. The server being shared among several users, the number of threads per process was capped in order to limit contention for CPU cores; the consequences of this sharing for the timing measurements are discussed in Chapter 4.

3.5 Evaluation Metrics

Let TP, TN, FP and FN denote respectively the numbers of true positives, true negatives, false positives and false negatives. For segmentation these quantities are counted at the pixel level over the tumor class, the metrics being computed per image and then averaged over the test set; for classification they are counted at the image level for each category.
The evaluation follows the philosophy of the original study, in which region-overlap quality remains the central criterion and is reported through Dice and normalised surface distance across heterogeneous tasks [1]. In the present BRISC setting, Dice and IoU are retained as the primary overlap measures, complemented by precision, recall and specificity for the profiling of errors at the pixel level.

3.5.1 Segmentation Metrics

Five complementary quantities are used to characterise segmentation performance, each capturing a different aspect of the agreement between the predicted mask and the reference annotation: overall overlap, the stricter union-based variant of that overlap, and the tendency of the model to over- or under-segment the lesion.

- **Dice similarity coefficient**, measuring the overlap between the predicted mask P and the reference mask G:

$$\text{Dice} = \frac{2\,|P \cap G|}{|P| + |G|} = \frac{2\,TP}{2\,TP + FP + FN}. \tag{3.5.1}$$

- **Intersection over union**, evaluating the same overlap relative to the union of the two regions:

$$\text{IoU} = \frac{|P \cap G|}{|P \cup G|} = \frac{TP}{TP + FP + FN}. \tag{3.5.2}$$

  The two quantities are monotonically related, but IoU penalises disagreement more severely and is systematically the lower of the two.

- **Precision**, the proportion of predicted tumor pixels that are correct, which measures over-segmentation:

$$\text{Precision} = \frac{TP}{TP + FP}. \tag{3.5.3}$$

- **Recall**, or sensitivity, the proportion of true tumor pixels that are retrieved, which measures under-segmentation:

$$\text{Recall} = \frac{TP}{TP + FN}. \tag{3.5.4}$$

  Reporting both is indispensable, since Dice alone cannot separate a model that overshoots the lesion boundary from one that misses part of it.

- **Specificity**, the proportion of background pixels correctly identified:

$$\text{Specificity} = \frac{TN}{TN + FP}. \tag{3.5.5}$$

Pixel accuracy is deliberately excluded from the headline metrics. The background overwhelmingly dominates a brain MRI slice, to the point that a model predicting an empty mask would still exceed 99% accuracy, a limitation already emphasised in the segmentation literature [13].

3.5.2 Classification Metrics

The classification branch is evaluated with a complementary set of quantities, chosen to remain informative even when the four BRISC categories are not perfectly balanced.

- **Accuracy**, the proportion of correctly classified images over the test set:

$$\text{Accuracy} = \frac{\text{correct predictions}}{\text{total predictions}}. \tag{3.5.6}$$

- **Precision** and **recall**, computed per class, the former expressing the reliability of a predicted label and the latter the coverage of an actual class.
- **F1-score**, their harmonic mean:

$$F_1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{3.5.7}$$

  The macro-averaged F1-score, obtained by averaging the per-class values with equal weight, is reported alongside accuracy so that the performance on the smaller categories is not concealed by the majority ones. The balanced accuracy, defined as the mean of the per-class recalls, is reported for the same reason.

3.6 Experimental Results

All the results reported here are obtained on the predefined BRISC test partitions, which were used neither for training, nor for validation, nor for model selection.

3.6.1 Segmentation Performance

The framework is first assessed on its primary objective, the delineation of the tumor region, using the five metrics introduced in Section 3.5. Table 3.4 reports the performance of the proposed framework on the 860 test images of the segmentation subset.

Table 3.4: Segmentation performance of the proposed framework on the BRISC test set.

Metric	**Value**
Dice Similarity Coefficient	0.9506
Intersection over Union	0.9078
Precision	0.9539
Recall	0.9491
Specificity	0.9993

The relation between precision and recall is the informative part of this table. At 0.9539 against 0.9491 the two are almost equal, with a slight advantage to precision, which indicates

that the predicted masks are marginally conservative: the model tends to omit a few boundary pixels rather than to extend beyond the lesion. For a prompt-driven method this is the preferable direction of error, since the bounding box already confines the prediction to the neighbourhood of the lesion, and an over-segmenting model would spill precisely into the healthy tissue that the box necessarily encloses. The specificity of 0.9993 confirms that the background is almost never claimed, although the value must be read in the light of the class imbalance discussed in Section 3.5.

3.6.2 Classification Performance

Attention now turns to the classification branch, which shares the same frozen image representation as the segmentation branch but is trained under a distinct objective. Table 3.5 reports the performance over the 1,000 test images of the four-class task.

Table 3.5: Classification performance of the MLP head on the BRISC test set.

Metric	**Value**
Accuracy	0.8430
Macro F1-score	0.8478
Balanced accuracy	0.8609

Two elements deserve comment. The balanced accuracy exceeds the raw accuracy by 1.8 points, which indicates that the errors are not concentrated on a single under-represented category but distributed over the four classes. And this level of performance is reached with a head applied to a representation that had to be computed for segmentation in any case, on top of a *frozen* encoder: the head has no means of reshaping that representation towards the classification objective and can only exploit the features that the segmentation-oriented fine-tuning happened to produce. The figures therefore characterise the shared representation under a fixed backbone, not the ceiling of the classification branch.

3.6.3 Comparison with Retrained Architectures

In order to situate the proposed framework, a set of representative segmentation architectures was retrained on BRISC under strictly identical conditions: same training and test partitions, same augmentation, same number of epochs, same evaluation procedure. This unified protocol is what confers meaning on the comparison, since it removes the differences in data handling and training budget that ordinarily separate published figures. Table 3.6 reports a representative selection of the outcome; the complete list of the twelve retrained baselines, with precision and recall, appears in Annex 4.7.

Three observations follow. The proposed framework leads by a wide margin: its Dice coefficient exceeds that of the best competing architecture, SABERNet, by 6.25 points, and the twelve retrained baselines as a whole occupy a band between 0.777 and 0.888 (Annex 4.7) that the framework never approaches. The internal ordering of that band is itself informative, attention-enhanced and Transformer-based designs occupying its upper part and purely convolutional ones its lower part, which reproduces the progression described in Chapter 2 and indicates that the

Table 3.6: Comparison of the proposed framework with representative segmentation architectures.

Architecture	Family	Dice	IoU
Proposed framework	**Foundation model**	**0.9506**	**0.9078**
MedSAM zero-shot	Foundation model	0.9005	0.8291
SABERNet	Attention	0.8881	0.8123
SegFormer-UAttNet	Transformer + CNN	0.8840	0.8067
U-Net	CNN	0.8801	0.7975
DeepLabV3+	CNN	0.7970	0.6920

retraining protocol did not distort the relative standing of the families.
The second observation concerns the *unadapted* model. Applied zero-shot, without a single gradient step on BRISC, MedSAM reaches a Dice of 0.9005, above SABERNet and more than twelve points above DeepLabV3+. The advantage does not originate in the fine-tuning procedure, then, but in the representation itself, since the model enters the task carrying features estimated from more than a million medical image–mask pairs [1] whereas the baselines are trained on the BRISC training subset alone.

The third observation attributes the remaining gap to the contributions of this chapter. Fine-tuning raises the zero-shot 0.9005 to 0.9390, and multi-bounding-box prompting brings the final configuration to 0.9506. The complete progression is reported in the first five rows of Annex 4.7, which constitute the ablation of the method, the subsequent rows constituting the full benchmark of which Table 3.6 reports a representative subset. It should be noted that the values reported here are those obtained under the present protocol and consequently differ from the figures published by the respective authors and reviewed in Chapter 2; they support a controlled comparison rather than a reproduction of published results.

3.6.4 Comparison with the Published State of the Art

The reference result on the BRISC segmentation benchmark is that of SegFormer-UAttNet, with a Dice coefficient of 0.9480 and an IoU of 0.9030 [15]. The proposed framework reaches 0.9506 and 0.9078 respectively, an improvement of 0.0026 in Dice and 0.0048 in IoU.
The margin is attributable to the conjunction of the three elements introduced in this chapter rather than to any one of them in isolation. The first is the recourse to a pretrained medical foundation model, whose representation is estimated on a corpus incommensurable with BRISC, where the competing architectures start from ImageNet weights or from scratch. The second is full fine-tuning, which lets that representation specialise to contrast-enhanced T1-weighted brain MRI instead of confining the adaptation to the decoder. The third is multi-bounding-box prompting, which supplies a spatial guidance proportional to the geometric complexity of the lesion.

3.7 Discussion

The benchmark of Section 3.6.3 points to a conclusion beyond the ranking itself: on a dataset of this size, the prior knowledge carried by the initialisation matters more than architectural refinement. No attention mechanism or hybrid Transformer–CNN design among the retrained baselines compensates for the absence of large-scale medical pretraining, since a model that has never seen a single BRISC image already surpasses all of them. Full fine-tuning then converts this generic prior into a domain-specific one, producing the second largest share of the total gain.

The prompting strategy's contribution should be read as conditional rather than general: separate boxes localise a fragmented or non-convex lesion more precisely than one enclosing rectangle, but for a compact lesion the strategy reduces to a single box and becomes indistinguishable from the original protocol. Its benefit therefore depends on the morphological composition of the dataset and would diminish on a cohort of simpler, single-component tumors.

A ceiling effect also qualifies the segmentation figures. The Dice coefficient of 0.9506 exceeds the inter-annotator agreement of 0.924 reported on BRISC [4]: the model now disagrees with the reference masks less than two human experts disagree with each other. Much of the residual gap is therefore more plausibly attributable to annotation variability than to model deficiency, making further gains against these masks hard to interpret.

The comparison with the published state of the art calls for the same caution. The reference architecture is fully automatic, receiving only the image, while the proposed framework additionally receives bounding-box prompts. A margin obtained under a strictly more informative input is better read as comparable performance through a different paradigm than as an advance over the state of the art — an asymmetry already flagged in Chapter 2 as a recurring weakness in how foundation models are evaluated.

Finally, the classification branch shows that a second task can be grafted onto the shared representation at negligible architectural cost. Its value here is less about accuracy than experimental control: with the encoder frozen and the head the only trainable component, any change in classification metrics can be attributed to the head alone — precisely the controlled baseline Chapter 4 requires.

3.8 Conclusion

This chapter presented a MedSAM-based framework for brain tumor analysis on BRISC, combining two-stage full fine-tuning, multi-bounding-box prompting, and a lightweight classification head. It reaches a Dice of 0.9506 (IoU 0.9078)$insegmentationand84.30\%$ accuracy in classification with fewer than 34,000 head parameters. This configuration serves as the reference for the next chapter, which integrates quantum components into the classification branch.

Chapter 4

Quantum Integration and Experimental Analysis

4.1 Introduction

This chapter presents the integration of a variational quantum circuit (VQC) into the classification head of the proposed brain tumor segmentation and classification framework. Rather than modifying the segmentation pathway, the quantum component is introduced specifically as an alternative to the classical multi-layer perceptron (MLP) used for classification.

The experiments focus on two main questions: how the quantum classification head performs compared with the classical MLP, and how its performance changes with different numbers of qubits and data-embedding strategies. Based on this ablation study, the most effective configuration is identified and used to define the final quantum classification architecture. The chapter concludes with a discussion of the observed performance, parameter efficiency, computational cost, and limitations of the proposed quantum integration.

4.2 Quantum Computing Background

This section introduces the notions on which the proposed component rests. No prior familiarity with quantum computing is assumed: the elements of Section 4.2.1 are sufficient to follow the rest of the chapter, and each is introduced with its classical counterpart wherever one exists. Section 4.2.2 then presents variational quantum circuits, which are best understood as small neural networks whose weights happen to be rotation angles.

One notational convention is used throughout. A quantum state is written between the symbols | and ⟩, as in $|\psi\rangle$; this is a conventional way of denoting a column vector and carries no meaning beyond that. The label inside the symbols names the state, so $|0\rangle$ and $|1\rangle$ denote the two reference states of a single qubit, playing the role of the values 0 and 1 of a classical bit.

4.2.1 Fundamentals of Quantum Computing

Before introducing these notions individually, it is useful to situate them within the broader landscape of computation. A classical computer manipulates bits that are always in one of two definite states, and a classical circuit is an ordered sequence of logical operations applied to those bits. A quantum computer instead manipulates qubits, which can exist in superposition and become entangled with one another, and a quantum circuit is the corresponding ordered sequence of operations applied to a register of qubits [25]. Because superposition and entanglement have no classical analogue, the set of operations available to a quantum circuit is considerably richer than the Boolean logic gates of a classical one, and this richness is what allows certain computational problems to be approached differently rather than merely faster. This paradigm has attracted interest well beyond physics, with proposed applications spanning cryptography, combinatorial optimisation, scientific simulation and, increasingly, machine learning; the present work is concerned with the last of these directions, where a small quantum circuit is trained jointly with a classical neural network as a compact component of a larger deep learning pipeline.

Four notions are sufficient here: the qubit, superposition, quantum gates and measurement.

Qubits and Superposition

The elementary unit of quantum information is the quantum bit, or *qubit*. Whereas a classical bit takes one of two values, a qubit is described by a normalized vector in a two-dimensional complex space,

$$|\psi\rangle = \alpha\,|0\rangle + \beta\,|1\rangle, \qquad \alpha, \beta \in \mathbb{C}, \qquad |\alpha|^2 + |\beta|^2 = 1. \tag{4.2.1}$$

The coefficients α and β are called *amplitudes*, and the constraint $|\alpha|^2 + |\beta|^2 = 1$ states that the vector has unit length. A qubit whose two amplitudes are non-zero is said to be in *superposition*. The term is easily misread: superposition does not mean that the qubit is secretly 0 or 1 and that we merely do not know which. The state is perfectly well defined; it is simply described by two numbers rather than by one binary value.

For a register of several qubits, the individual states are combined by the tensor product, denoted $\otimes$, into a single multi-qubit state; a register of n_q qubits initialized in $|0\rangle$ is therefore written $|0\rangle^{\otimes n_q}$, which for $n_q = 4$ means $|0\rangle \otimes |0\rangle \otimes |0\rangle \otimes |0\rangle = |0000\rangle$ [7]. Such a register is described not by n_q pairs of numbers but by 2^{n_q} amplitudes taken together: four qubits require 16, eight require 256. Simulating a quantum circuit on an ordinary computer means storing and updating all of these numbers explicitly, so the cost of simulation grows exponentially with the number of qubits. This single fact governs the range of circuit widths explored in this chapter.

Quantum Gates

Operations on a quantum register are called *gates*, by analogy with the logic gates of a classical circuit, and each one is a matrix that multiplies the state vector. These matrices are *unitary*, which means that they preserve the length of the vector: since the amplitudes must always satisfy $|\alpha|^2 + |\beta|^2 = 1$, no operation may change that normalization, and every quantum operation is consequently reversible.

Gates fall into two families according to whether they carry trainable parameters [2]. *Fixed* gates perform predetermined transformations that never change during training, such as the Hadamard gate, which turns a basis state into an even superposition of both basis states, or the CNOT gate, which establishes entanglement. *Parameterized* gates depend on adjustable values, usually rotation angles such as R_X or R_Y, and constitute the trainable degrees of freedom of the model. The gates relevant to this work are collected in Table 4.1.

The role of the CNOT gate deserves to be made concrete, since it is what distinguishes a quantum layer from a collection of independent single-qubit operations. Consider a control qubit prepared in the superposition $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ and a target qubit in $|0\rangle$. Their joint state is $\frac{1}{\sqrt{2}}(|00\rangle + |10\rangle)$, which can still be written as a product of two independent qubits. Applying the CNOT gate gives

$$\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle), \tag{4.2.2}$$

Table 4.1: Quantum gates used in this work: unitary matrices and effect [2].

Gate	**Qubits**	**Unitary matrix**	**Effect**
Pauli-X	1	$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$	Analogous to the classical NOT gate: switches $\|0\rangle$ to $\|1\rangle$ and conversely.
Pauli-Z	1	$\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$	Rotation through π radians about the Z axis; used here as the measured observable.
$R_Y(\theta)$	1	$\begin{pmatrix} \cos\frac{\theta}{2} & -\sin\frac{\theta}{2} \\ \sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{pmatrix}$	Rotates the state vector about the Y axis by θ; carries the angle embedding.
$R_Z(\theta)$	1	$\begin{pmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{pmatrix}$	Rotates the state vector about the Z axis by θ.
Hadamard	1	$\frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$	Transforms a basis state into an even superposition of the two basis states.
$R(\phi,\theta,\omega)$	1	$\begin{pmatrix} e^{-i(\phi+\omega)/2}\cos\frac{\theta}{2} & -e^{i(\phi-\omega)/2}\sin\frac{\theta}{2} \\ e^{-i(\phi-\omega)/2}\sin\frac{\theta}{2} & e^{i(\phi+\omega)/2}\cos\frac{\theta}{2} \end{pmatrix}$	General rotation combining $R_Z(\omega)R_Y(\theta)R_Z(\phi)$; the trainable gate of the proposed circuit.
CNOT	2	$\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$	Applies Pauli-X to the target qubit if the control qubit is $\|1\rangle$; creates entanglement.

a state that admits no such factorization: the two qubits are now *entangled*, and the outcome of measuring one is fully correlated with the outcome of measuring the other [2]. When the qubits of a register are linked by CNOT gates, a change affecting one qubit propagates to the others, which is what allows a shallow circuit to represent relationships among its inputs rather than treating each input in isolation.

Measurement

The amplitudes of a quantum state cannot be read out. Measurement is fundamentally different from classical observation: it forces the system to collapse from a superposition onto one of its basis states, the projection being probabilistic with probabilities determined by the state [7]. For a single qubit in the state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, the probabilities of observing each basis state are

$$P(0) = |\alpha|^2, \qquad P(1) = |\beta|^2. \tag{4.2.3}$$

The most common measurement scheme uses the Pauli operators, and in particular the Pauli-Z operator, whose eigenvalues $+1$ and -1 indicate whether the qubit is more likely to be found in $|0\rangle$ or in $|1\rangle$ [7]. Rather than single-shot outcomes, which are binary and stochastic, variational algorithms work with *expectation values*. Expanding $\langle\psi|Z|\psi\rangle$ gives

$$\langle Z \rangle = |\alpha|^2 - |\beta|^2 = P(0) - P(1) \in [-1, 1], \tag{4.2.4}$$

so the measured value is the difference between the probabilities of the two basis states [2]. Its practical importance is that it is a continuous number rather than a random bit: it varies smoothly with the circuit parameters, which is exactly what a gradient-based optimizer requires. The measurement stage therefore acts as the mapping $M : |\psi_{\text{out}}\rangle \to y$ that converts the quantum state back into classical values usable by the following layers [7].

4.2.2 Variational Quantum Circuits

Variational quantum circuits (VQCs), also called parameterized quantum circuits, are the building blocks of quantum neural networks. They consist of quantum circuits with tunable parameters that can be optimized like the weights of a classical network, adjusted iteratively to minimize a cost function, which is what enables the model to learn from data [2, 7]. A circuit alternates layers of parameterized rotation gates with entangling layers built from CNOT gates.

Formally, a circuit is a sequence of unitary transformations parameterized by a set of angles $\theta = \{\theta_1, \ldots, \theta_L\}$,

$$U(\theta) = U_L(\theta_L) \cdots U_2(\theta_2)\, U_1(\theta_1), \tag{4.2.5}$$

where $U_i(\theta_i)$ is the transformation applied at the i-th layer and L is the number of layers [2]. The optimization is hybrid: a classical optimizer updates the angles on the basis of a cost function evaluated through measurement of the circuit.

Viewed as a function acting on classical data, a VQC decomposes into three successive stages [7]: an embedding E mapping the classical vector to a quantum state, one or more parameterized layers L transforming that state, and a measurement M recovering a classical output. The circuit is their composition,

$$Q(x) = (M \circ L \circ E)(x), \tag{4.2.6}$$

where $\circ$ denotes function composition. Read in this way, a variational circuit takes a vector of numbers and returns a vector of numbers, exactly like a small neural network, and is trained in the same manner. Table 4.2 makes the correspondence explicit, since the remainder of the chapter uses the vocabulary of neural networks to describe the circuit.

Data Embedding

Since quantum circuits operate exclusively on quantum states, classical data must be encoded into such a state before any parameterized layer can be applied, and this encoding determines how the information is mapped onto the underlying space [2]. The register is initialized in $|0\rangle^{\otimes n_q}$ and a unitary operation $U(x)$ maps it onto a data-dependent state:

$$E(x) = U(x)\, |0\rangle^{\otimes n_q} = |x\rangle^{\otimes n_q}. \tag{4.2.7}$$

Table 4.2: Correspondence between the stages of a variational quantum circuit and those of a classical neural network.

In the circuit	**Role**	**Classical counterpart**
Data embedding E	Brings the input values into the circuit	Input layer
Parameterized layers L	Trainable transformation of the state	Hidden layers
Rotation angles θ	The values adjusted during training	Network weights
Entangling gates	Couple the qubits to one another	Connections between units
Measurement M	Returns the numbers used downstream	Output layer

Several strategies exist [2, 7]. *Basis encoding* writes classical bits directly into qubit states. *Amplitude encoding* stores a vector $x = [x_1, \ldots, x_N]$ in the probability amplitudes,

$$|\psi\rangle = \sum_{i=0}^{2^{n_q}-1} \frac{x_i}{\|x\|} |i\rangle, \qquad n_q = \lceil \log_2 N \rceil, \tag{4.2.8}$$

which is compact, since the qubit count grows only logarithmically with the size of the vector, but requires a global normalization by the Euclidean norm $\|x\|$. *Hamiltonian-based embeddings* exploit problem-specific Hamiltonians. Finally, *angle encoding* assigns one qubit per feature and uses the feature value, after scaling, as the rotation angle of a single-qubit gate:

$$|\psi(x)\rangle = \bigotimes_{j=1}^{n_q} R_Y(x_j) |0\rangle = \bigotimes_{j=1}^{n_q} \left[\cos\left(\frac{x_j}{2}\right) |0\rangle + \sin\left(\frac{x_j}{2}\right) |1\rangle \right], \qquad n_q = n. \tag{4.2.9}$$

Angle encoding requires a minimal number of qubits and few gates, and is hardware-efficient [7]; it needs no global normalization, so each feature is mapped independently of the magnitude of the others [2]. Its qubit count grows linearly with the number of features, so the classical network must first compress its representation to as many values as there are qubits, as described in Section 4.3.2. One consequence is central to this chapter: with angle encoding the number of qubits *is* the width of the interface between the classical network and the circuit, so varying it varies how much information reaches the quantum block. The choice between the two encodings is examined experimentally in Section 4.5.1.

Parameterized Layers and Data Re-Uploading

Once the data are embedded, the circuit applies one or more quantum layers. A generic layer L is defined by a trainable unitary $U(w)$, where w denotes the circuit weights; acting on an input state $|X_{n_q}\rangle$ it produces [7]

$$L : U(w) |X_{n_q}\rangle = |Y_{n_q}\rangle. \tag{4.2.10}$$

A recognized limitation of the basic scheme is that encoding the data once, at the input, bounds the class of functions the circuit can represent. *Data re-uploading* addresses this by re-

applying the embedding at every layer, so that each qubit sees the data several times interleaved with trainable rotations, which increases expressivity without increasing the number of qubits [26]. This mechanism is available to angle encoding, which is a sequence of ordinary rotation gates, but not to amplitude encoding, which is a state preparation and must be the first operation of the circuit. This is the scheme adopted in Section 4.3.3.

4.3 Proposed Quantum Classification Head

This section presents the proposed quantum classification head, covering its placement within the architecture, the preparation of the classical features, the circuit structure, and the final classification step.

4.3.1 Placement of the Quantum Circuit

The quantum circuit is introduced only in the classification branch of the proposed MedSAM architecture, as illustrated in Figure 4.1. The segmentation branch remains unchanged.

After the image encoder, the extracted features are used by two branches. The first branch performs tumor segmentation, while the second branch performs image classification. In the classification branch, global average pooling converts the spatial feature representation into a 256-dimensional feature vector.

The classical MLP previously used for classification is then replaced by the proposed quantum classification head. The quantum circuit is therefore not used for image segmentation or for feature extraction; its role is specifically to transform the compact image representation into a four-class classification decision. This position allows the quantum component to operate on a small and meaningful representation rather than directly on the high-dimensional feature map produced by the encoder.

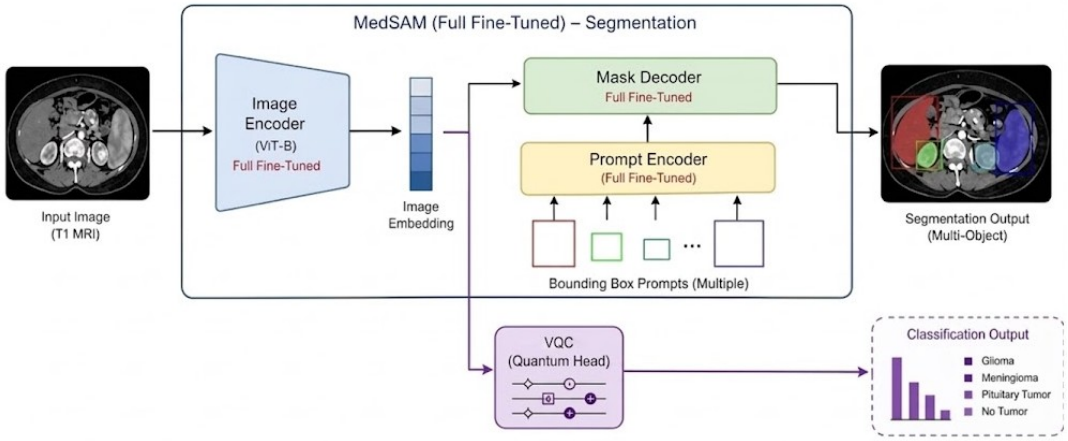


Figure 4.1: Proposed MedSAM architecture with quantum classification head for joint segmentation and classification.

4.3.2 Preparation of Classical Features

The MedSAM encoder produces a high-dimensional feature representation. After global average pooling, this representation is reduced to a vector of 256 features.

For angle embedding, the number of values entering the circuit must correspond directly to the number of qubits, since each qubit carries exactly one rotation angle. A classical linear layer therefore projects the 256-dimensional feature vector to n_q values, where n_q denotes the number of qubits. The resulting values are scaled to an appropriate range and used as rotation angles, so that each classical feature is associated with one qubit. For example, with eight qubits, the 256-dimensional pooled representation is reduced to eight values before entering the circuit.

Amplitude embedding follows a different constraint: a register of n_q qubits can hold a state vector of dimension 2^{n_q}, so the classical representation must be projected (or, when $256 = 2^{n_q}$, used directly) to that dimension rather than to n_q raw values. In the amplitude-embedding configuration evaluated in Section 4.5.1, the pooled representation was projected to 256 dimensions before normalization and state preparation, corresponding to $n_q = 8$ qubits. This distinction matters for the interpretation of Section 4.5.1: an amplitude circuit narrower than the one implied by the "eight qubits" label used for comparison introduces an additional compression step, on top of the encoding itself.

4.3.3 Quantum Circuit Architecture

The proposed quantum classification head is implemented as a variational quantum circuit (VQC) composed of three layers. Each layer follows the same sequence of operations: angle embedding, trainable rotations, and entangling gates, as illustrated in Figure 4.2. The circuit is built from five successive steps, described below in order; to make each one concrete, every step is accompanied by a small illustration on two qubits, worked out by hand rather than left as an abstract formula.

1. **Angle embedding :** At the beginning of the first layer, the classical input features are encoded into the eight qubits using R_Y rotations. Each feature is associated with one qubit, such that x_i is encoded through an $R_Y(x_i)$ rotation on qubit q_i. This produces the initial quantum representation of the input data.

   Two-qubit illustration. Take two qubits, initialised in $|00\rangle$, and two classical values $x_0 = \pi/2$ and $x_1 = 0$. Using $R_Y(\theta)|0\rangle = \cos\frac{\theta}{2}|0\rangle + \sin\frac{\theta}{2}|1\rangle$, embedding each qubit individually gives

   $$R_Y(x_0)|0\rangle = \tfrac{1}{\sqrt{2}}\big(|0\rangle + |1\rangle\big), \tag{4.3.1}$$

   $$R_Y(x_1)|0\rangle = |0\rangle, \tag{4.3.2}$$

   so that embedding both qubits together gives

   $$\big(R_Y(x_0) \otimes R_Y(x_1)\big)|00\rangle = \tfrac{1}{\sqrt{2}}\big(|00\rangle + |10\rangle\big). \tag{4.3.3}$$

   Only q_0 carries any superposition here, since $x_1 = 0$ leaves q_1 unrotated; this is exactly

what it means for each qubit to encode its own feature independently at this stage.

2. **Trainable rotation :** The encoded quantum state is then processed by trainable general rotations. Each rotation contains learnable parameters that are optimized during training.

   Two-qubit illustration. Consider the restricted case $R(\phi, \theta, \omega)$ with $\phi = \omega = 0$, so that the trainable gate reduces to a further $R_Y(\theta)$ rotation about the same axis as the embedding. Applying $R_Y(\pi/4)$ to q_0 on top of the embedding in Equation (4.3.1) simply adds the two angles, since both rotations act about the Y axis:

   $$R_Y(\pi/4)\, R_Y(\pi/2)|0\rangle = R_Y(3\pi/4)|0\rangle = \cos\tfrac{3\pi}{8}|0\rangle + \sin\tfrac{3\pi}{8}|1\rangle. \tag{4.3.4}$$

   This special case makes the role of the trainable gate visible without extra machinery: it further rotates whatever the embedding produced, rather than replacing it. In the general case $\phi, \omega \neq 0$, the gate additionally applies R_Z rotations before and after $R_Y(\theta)$; these do not change the probabilities $P(0), P(1)$ of a single qubit measured in isolation, but they do change the relative phase between $|0\rangle$ and $|1\rangle$, which becomes observable once the qubit is entangled with another through a CNOT gate. For the remainder of this illustration the trainable gate is set to identity, so that the effect of entanglement and re-uploading can be isolated in turn.

3. **Entanglement :** After these rotations, CNOT gates are applied to establish interactions between the qubits. The CNOT gates follow a ring topology, where each qubit interacts with its neighbouring qubit, with the last qubit connected back to the first one. This allows information to be exchanged between the different qubits rather than processing them independently.

   Two-qubit illustration Returning to the embedded state of Equation (4.3.3) and applying a CNOT gate with q_0 as control and q_1 as target,

   $$|\psi_1\rangle = \mathrm{CNOT}\Big[\tfrac{1}{\sqrt{2}}\big(|00\rangle + |10\rangle\big)\Big] = \tfrac{1}{\sqrt{2}}\big(|00\rangle + |11\rangle\big). \tag{4.3.5}$$

   This is a Bell state. It can no longer be written as a state for q_0 combined with a separate state for q_1 — the two qubits are now entangled, which is precisely what the CNOT gate was introduced to achieve.

4. **Data re-uploading :** After the first layer, the resulting quantum state is not reset. It is directly passed to the second layer. At the same time, the original classical input features are encoded again using new $R_Y(x_i)$ rotations on the same eight qubits. Therefore, the angle embedding in the second layer does not encode the previous quantum state; instead, it re-encodes the classical input data while acting on the quantum state inherited from the previous layer. The same process is repeated in the third layer. This mechanism is referred to as *data re-uploading*: it allows the same classical information to interact repeatedly with the trainable quantum operations while keeping the number of qubits fixed.

   Two-qubit illustration The same values $x_0 = \pi/2, x_1 = 0$ are embedded a second time, but now applied to the state $|\psi_1\rangle$ of Equation (4.3.5) rather than to $|00\rangle$. Expanding term by

term, and using $R_Y(\theta)|1\rangle = -\sin\frac{\theta}{2}|0\rangle + \cos\frac{\theta}{2}|1\rangle$,

$$\big(R_Y(x_0) \otimes R_Y(x_1)\big)|\psi_1\rangle = \tfrac{1}{2}\big(|00\rangle - |01\rangle + |10\rangle + |11\rangle\big). \tag{4.3.6}$$

Before this second embedding, only $|00\rangle$ and $|11\rangle$ carried any amplitude; afterwards, all four basis states do. Nothing here was recomputed from $|\psi_1\rangle$: the angles x_0, x_1 are the same numbers as in layer 1, and the operator applied is the same fixed $R_Y(x_0) \otimes R_Y(x_1)$. What changed is the state it acted on — the same rotation redistributes amplitude differently once the register is already entangled.

5. **Measurement :** After the third layer, the final quantum state is passed to the measurement stage, where two types of observables are evaluated. The Pauli-Z observable is measured on each of the eight qubits, producing eight values. In addition, Pauli-$Z \otimes Z$ observables are measured between neighbouring qubits, including the connection between the last and first qubits, producing another eight values. The quantum circuit therefore provides a total of 16 measurement values to the final classical linear layer.

   Two-qubit illustration Consider the Bell state of Equation (4.3.5), $|\psi_1\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$, which assigns probability $\frac{1}{2}$ to $|00\rangle$ and $\frac{1}{2}$ to $|11\rangle$. Since $Z|0\rangle = +|0\rangle$ and $Z|1\rangle = -|1\rangle$, the two single-qubit expectation values are

$$\langle Z_0\rangle = (+1)\cdot\tfrac{1}{2} + (-1)\cdot\tfrac{1}{2} = 0, \tag{4.3.7}$$

$$\langle Z_1\rangle = (+1)\cdot\tfrac{1}{2} + (-1)\cdot\tfrac{1}{2} = 0, \tag{4.3.8}$$

   since q_0 (and equally q_1) is 0 half the time and 1 half the time. Taken alone, Equations (4.3.7)–(4.3.8) report nothing: they cannot distinguish this entangled state from a qubit in an equal classical mixture of 0 and 1. The correlator tells a different story:

$$\langle Z_0 Z_1\rangle = (+1)(+1)\cdot\tfrac{1}{2} + (-1)(-1)\cdot\tfrac{1}{2} = 1. \tag{4.3.9}$$

   since the two qubits are always found in the *same* basis state. This single example makes concrete why the proposed readout measures $Z \otimes Z$ correlators in addition to individual Z values: for an entangled state, the individual qubit measurements can be entirely uninformative while the pairwise correlator in Equation (4.3.9) carries the information the circuit's entangling gates were built to produce.

In the final configuration, the circuit uses eight qubits and three layers. The overall structure can therefore be summarized as

$$\text{Input} \to [R_Y(x) \to \text{Trainable Rotations} \to \text{CNOT Ring}]^{\times 3}\,.$$

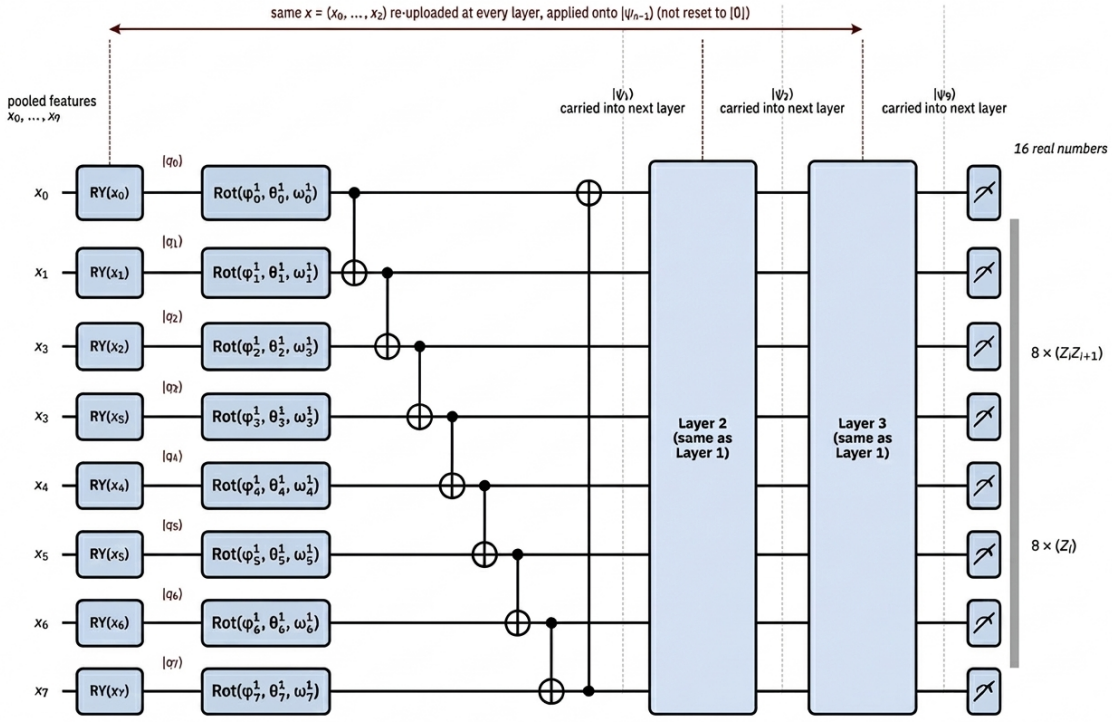


Figure 4.2: Architecture of the proposed eight-qubit, three-layer variational quantum classification head employing data re-uploading and Z/ZZ measurements.

4.3.4 Quantum-to-Classical Classification

After the three quantum layers, the final quantum state must be converted back into classical values in order to perform the classification.

For this purpose, the circuit uses measurements based on the Pauli-Z operator. A Z measurement is performed on each qubit to obtain information about its final state. In addition, measurements between neighbouring qubits are used to capture information about their interactions.

For n_q qubits, this produces $2n_q$ classical values. With the final configuration of eight qubits, the quantum circuit therefore produces 16 values. These values are passed directly to a classical linear layer, which produces four output values corresponding to the four BRISC classes: glioma, meningioma, pituitary tumor, and no tumor. The class associated with the highest output value is selected as the final prediction.

4.4 Experimental Setup

The experiments are designed to evaluate the proposed quantum classification head under controlled conditions. The classical MLP used in Chapter 3 serves as the reference, while the quantum head is evaluated with different data embeddings and numbers of qubits. The same general training protocol is used for all configurations.

- **Controlled classification-head comparison.** All experiments start from the same MedSAM checkpoint obtained in Chapter 3. The image encoder is kept frozen, so that

it produces the same image features for every experiment. Only the classification head is changed, either using the classical MLP or the proposed quantum head. This setup makes the comparison fair: any difference in classification performance comes from the classification head rather than from changes in the image representation. The segmentation branch is also kept unchanged; since the quantum component is used only for classification, the segmentation performance remains the same across the different classification heads.

- **Embedding comparison.** The first experiment compares two ways of encoding the classical features into the quantum circuit: angle embedding and amplitude embedding. The number of qubits is fixed to eight, and the other circuit components are kept unchanged, so that the only difference is how the classical features are introduced into the quantum circuit. This experiment is used to determine which embedding strategy is more suitable for the proposed classification task.

- **Qubit-width comparison.** The second experiment evaluates the effect of the number of qubits. Angle embedding is used in all cases, while the quantum circuit is tested with 4, 6 and 8 qubits, the number of layers being fixed to three. Increasing the number of qubits allows the circuit to process more input values and provides a larger quantum representation, so the objective is to determine whether increasing the circuit size improves classification performance.

- **Training configuration.** All experiments are trained using the same general optimization procedure, on the computational environment described in Section 3.6.4. The quantum circuits are implemented with the PennyLane library [27] and executed on a classical CPU using a statevector simulator with adjoint differentiation. The experiments therefore do not use a physical quantum computer; they simulate the behaviour of an ideal, noiseless quantum circuit. One exception applies: amplitude embedding is implemented as a state preparation, through which adjoint differentiation cannot propagate gradients, so that configuration is executed with a backpropagation-based simulator instead. The main training settings are summarized in Table 4.3.

4.5 Results

The experiments are designed to evaluate the proposed quantum classification head and to determine the configuration that provides the best balance between classification performance and model complexity. Three aspects are studied: the data embedding strategy, the number of qubits, and the number of variational layers.

All experiments use the same MedSAM backbone and the same classification protocol. They are evaluated on the predefined BRISC test partition, while the classical MLP is used as the reference model. Since the backbone is kept frozen, the comparison focuses specifically on the classification head.

The classification accuracy is used as the primary metric for comparing the different configurations. In addition, balanced accuracy and macro-F1 are considered to provide a more complete evaluation, particularly in the presence of class imbalance.

Table 4.3: Configuration used for the classification-head experiments.

Parameter	**Value**
Backbone	MedSAM checkpoint, frozen
Number of qubits	4, 6 or 8
Number of layers	1, 2 or 3
Encoding	Angle embedding (R_Y) or amplitude embedding
Trainable gate	$R(\phi, \theta, \omega)$ on each qubit
Entanglement	Ring of CNOT gates
Readout	$\langle Z_q \rangle$ and $\langle Z_q Z_{q+1} \rangle$
Optimizer	AdamW with weight decay 10^{-4}
Learning rate	5×10^{-4} (quantum circuit), 10^{-4} (classical layers)
Maximum epochs	50
Early stopping	Patience = 10, based on validation accuracy
Effective batch size	4
Simulation	PennyLane statevector simulator, CPU
Random seed	42

4.5.1 Effect of the Data Embedding

The first experiment investigates the effect of the data-encoding strategy used to introduce the classical features into the quantum circuit. Two approaches are compared: angle embedding and amplitude embedding. Both configurations use eight qubits and follow the same general classification-head architecture, while retaining the corresponding encoding strategy and circuit implementation. The results are presented in Table 4.4.

Table 4.4: Comparison between angle embedding and amplitude embedding using eight qubits.

Metric	**Angle embedding**	**Amplitude embedding**
Classification accuracy	**0.8540**	0.5930
Macro-F1	**0.8551**	0.6203
Balanced accuracy	**0.8723**	0.6219
Circuit input width	8 values	256 values
Head parameters	2,708	652
Throughput	1.6 img/s	3.3 img/s

A substantial difference is observed between the two encoding strategies. Angle embedding achieves a classification accuracy of 85.40%, compared with only 59.30% for amplitude embedding. The same trend is observed for Macro-F1 and balanced accuracy, with angle embedding reaching 85.51% and 87.23%, respectively, compared with 62.03% and 62.19% for amplitude embedding. Overall, angle embedding provides a considerably more effective representation for the considered classification task, with an accuracy improvement of approximately 26 percentage points.

The lower performance of amplitude embedding can be related to the way the classical representation is encoded into the quantum state. Amplitude embedding requires the input vector to be normalized before state preparation, meaning that the representation is encoded through

its relative amplitudes. Consequently, information associated with the original magnitude of the feature vector is not preserved directly. In addition, the implemented amplitude-embedding configuration does not use data re-uploading, whereas the angle-embedding circuit repeatedly introduces the classical features throughout the variational circuit. This provides the angle-based configuration with additional opportunities to transform and exploit the input representation.

Amplitude embedding nevertheless offers computational advantages. It uses fewer trainable parameters (652 compared with 2,708) and achieves a higher throughput of 3.3 images/s compared with 1.6 images/s for angle embedding. However, these efficiency gains come at the cost of a substantial degradation in classification performance. Therefore, computational efficiency alone does not justify selecting amplitude embedding for the considered task.

Based on these results, angle embedding is retained for the subsequent experiments. The experiment demonstrates that the data-encoding strategy is a major factor in the effectiveness of the quantum classification head, and that, for the considered architecture and dataset, angle embedding provides a substantially better trade-off between representation and classification performance.

4.5.2 Effect of the Number of Qubits

After selecting angle embedding, the second experiment studies the effect of the number of qubits. Three configurations are evaluated using 4, 6 and 8 qubits, the number of quantum layers being fixed to three for all configurations. The results are compared with the classical MLP and reported in Table 4.5.

Table 4.5: Comparison between the classical MLP and quantum classification heads with different numbers of qubits.

Metric	**MLP**	**Quantum 4q**	**Quantum 6q**	**Quantum 8q**
Classification accuracy	0.8430	0.8320	0.8380	**0.8540**
Macro-F1	0.8478	0.8318	0.8408	**0.8551**
Balanced accuracy	0.8609	0.8528	0.8581	**0.8723**
Accuracy difference vs. MLP	–	−1.10	−0.50	**+1.10**
Head parameters	33,924	**1,612**	2,160	2,708
Reduction vs. MLP	–	95.2%	93.6%	92.0%
Circuit parameters	–	36	54	72
Throughput	**6.0** img/s	2.8 img/s	2.2 img/s	1.5 img/s

The results show a gradual improvement as the number of qubits increases. The classification accuracy rises from 83.20% with four qubits to 83.80% with six qubits and reaches 85.40% with eight qubits, and the same ordering holds for Macro-F1 and balanced accuracy.

The eight-qubit configuration gives the best results for all three classification metrics, and is the only quantum configuration that exceeds the classical MLP in the reported experiment. This behaviour is consistent with the role of the qubit count under angle embedding: since each qubit

carries one input value, a wider register both transmits more information to the circuit and increases its capacity, which grows as $9n_q$ trainable angles over a state space of dimension 2^{n_q}. Increasing the number of qubits also increases the computational cost, as discussed in Section 4.5.5.

4.5.3 Effect of the Number of Variational Layers

Once the encoding and the register width had been fixed, a third ablation examined the depth of the circuit. The register was held at eight qubits with angle embedding, and the number of variational layers was varied from one to three. Every other element was left unchanged, including the dataset splits, the optimizer, the learning-rate schedule, the early-stopping rule and the random seed, so that the depth of the circuit was the only differing factor between the three arms. Table 4.6 reports the outcome.

Table 4.6: Effect of the number of variational layers on classification performance, at eight qubits with angle embedding.

Metric	**1 layer**	**2 layers**	**3 layers**
Accuracy	0.8420	0.8480	**0.8540**
Macro-F1	0.8470	0.8539	**0.8551**
Balanced accuracy	0.8597	0.8674	**0.8723**
Circuit parameters	24	48	72
Head parameters	2,660	2,684	2,708

The three metrics improve monotonically with depth. Accuracy rises by 1.20 percentage points between the shallowest and the deepest circuit, Macro-F1 by 0.81 and balanced accuracy by 1.26, and no metric contradicts the ordering. The behaviour is consistent with the role of depth in a re-uploading circuit: each additional layer re-encodes the same eight values and interleaves them with a further set of trainable rotations, so the family of functions the circuit can represent is enlarged without any widening of the interface to the classical network.

The parametric cost of this gain is modest. Depth adds $3n_q = 24$ angles per layer, so the circuit grows from 24 to 72 trainable parameters, and the complete head from 2,660 to 2,708, an increase of less than 2% of the head budget. The cost in training time is more substantial, the three-layer configuration requiring approximately twice as long per epoch as the one-layer configuration, since the simulator must execute three embedding and entangling blocks instead of one for every image.

Two reservations qualify this trend. The first is statistical: each configuration was trained once, with a single seed, and given the sample size of the test set, a gap of 1.20 percentage points falls within the range of variation expected from sampling alone. The ordering is consistent across the three metrics, which makes it more credible than an isolated difference would be, but it cannot be declared significant without replication over several seeds.

The second concerns the training budget. The one- and two-layer circuits reached their best validation performance at epochs 34 and 28 respectively, whereas the three-layer circuit reached its own at epoch 48, that is, within two epochs of the imposed limit of 50. The deepest arm was therefore still improving when training terminated, so its reported result is a lower bound on what the configuration can reach, while the shallower arms had plateaued.

The three-layer circuit was retained for the final configuration. It attains the best value on all three classification metrics, its advantage is coherent across metrics rather than confined to one, and it costs only 48 additional parameters relative to the shallowest arm. The final quantum classification head therefore comprises eight qubits, angle embedding with data re-uploading, and three variational layers.

4.5.4 Comparison with the Classical MLP

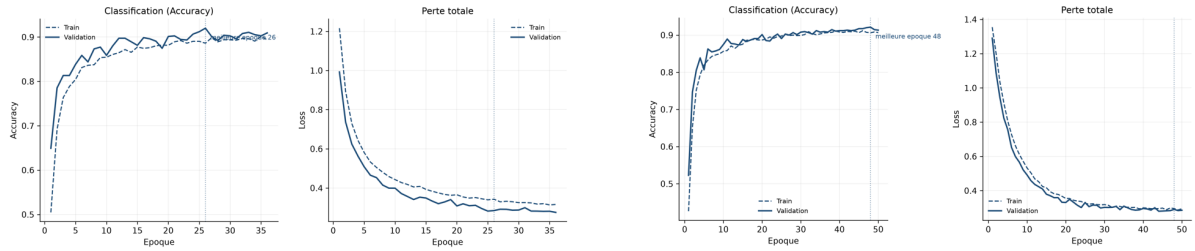
The final quantum configuration uses eight qubits, angle embedding and three variational layers. It is compared with the classical MLP used as the reference classification head in Chapter 3. To ensure a fair comparison, the MedSAM backbone, the segmentation pipeline, the training data and the evaluation procedure are kept unchanged, so that the classification head is the only difference between the two experiments.

As shown in Table 4.7, the quantum head achieves an accuracy of 85.40%, compared with 84.30% for the classical MLP. It also obtains higher Macro-F1 and balanced accuracy, with improvements of +0.73 and +1.14 percentage points respectively.

Table 4.7: Comparison between the classical MLP and the best-performing eight-qubit quantum classification head.

Metric	MLP	Quantum 8q	Improvement
Accuracy	0.8430	**0.8540**	+1.10 **pts**
Macro-F1	0.8478	**0.8551**	+0.73 **pts**
Balanced accuracy	0.8609	**0.8723**	+1.14 **pts**

The training behaviour of the two classification heads is shown in Figure 4.3. Only the classification accuracy and the classification loss are presented, since the purpose of this comparison is to evaluate the effect of replacing the classical MLP with the quantum head.



(a) Classical MLP (b) Quantum head

Figure 4.3: Training and validation curves of the classical MLP and the selected quantum classification head.

The two models show a similar overall training behaviour. In both cases, the classification accuracy increases during the early stages of training and then becomes relatively stable, while the classification loss decreases. The training and validation curves remain relatively close during the later epochs, which indicates stable optimization without a clear sign of overfitting.

The quantum head achieves slightly better classification performance than the classical MLP while using substantially fewer trainable parameters. It should nevertheless be noted that, as with the layer-depth ablation, each configuration was trained once with a single seed, and given the size of the test set, a gap of 1.10 percentage points falls within the range of variation expected from sampling alone. The two heads are therefore best described as achieving comparable classification performance in this experiment.

4.5.5 Parameter Efficiency and Computational Cost

One of the main advantages observed in the experiments is the reduction in the number of trainable parameters. The classical MLP contains 33,924 parameters, whereas the quantum heads contain between 1,612 and 2,708 parameters. The eight-qubit quantum head therefore uses 92.0% fewer parameters than the classical MLP, and the four-qubit configuration achieves an even larger reduction of 95.2%.

The quantum circuit itself contains only 36, 54 and 72 trainable parameters for four, six and eight qubits respectively. This shows that the quantum component provides a very compact alternative to the classical classification head. Unlike the differences observed in the classification metrics, this reduction is exact rather than statistical, and it was obtained in every configuration evaluated.

This reduction in parameters nevertheless comes with a computational cost. The MLP reaches a throughput of 6.0 images per second, while the throughput decreases to 2.8, 2.2 and 1.5 images per second for the four-, six- and eight-qubit configurations respectively. This additional cost is mainly related to the classical simulation of the quantum circuit, which must construct and differentiate the 2^{n_q} amplitudes of the register once per image. The proposed approach should therefore not be considered faster than the classical MLP in the current experimental setup; its main advantage is the strong reduction in the number of trainable parameters while maintaining comparable classification performance.

4.5.6 Qualitative Results

The quantitative tables report averages over the test set and say nothing about the behaviour of the framework on individual cases. Figure 4.4 therefore presents the output of the final configuration, combining the fully fine-tuned MedSAM segmentation branch with the eight-qubit quantum classification head, on one representative case of each of the four BRISC categories: glioma, meningioma, pituitary tumor and no tumor.

Three observations complement the numerical results. The predicted contours follow the reference annotation closely on the three tumor categories, which is consistent with the Dice

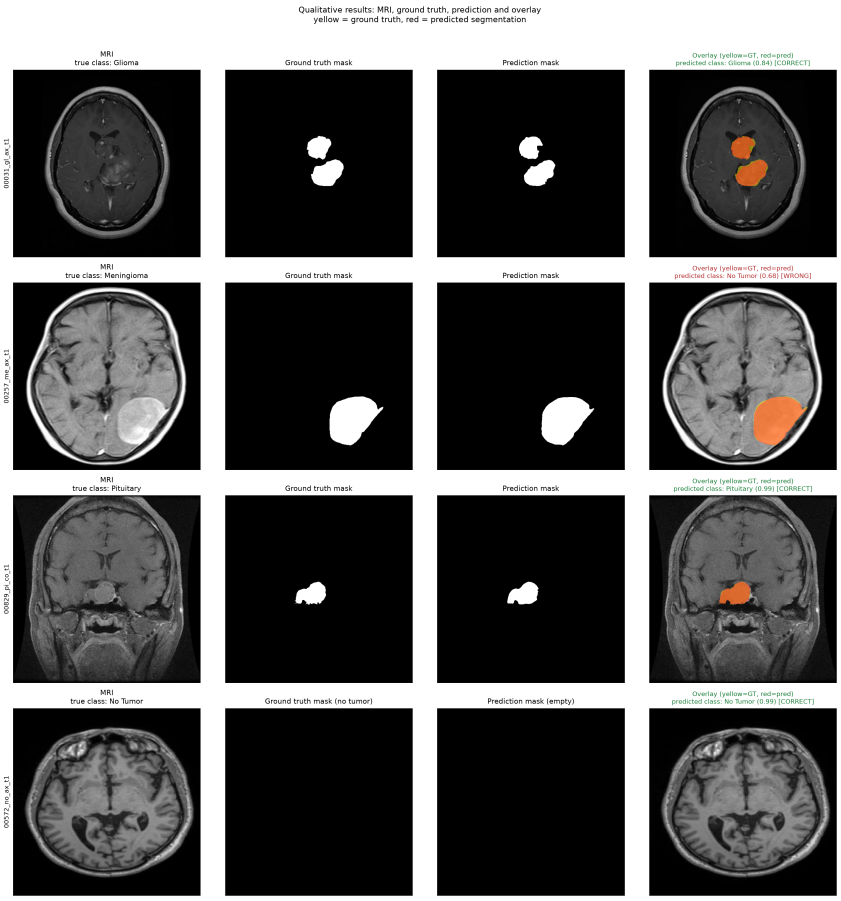


Figure 4.4: Qualitative results of the final framework on one representative case of each BRISC category.

coefficient reported in Chapter 3, and the residual disagreement is concentrated on the boundary rather than distributed over the region, which corresponds to the near-equality between precision and recall observed there. The three tumor types differ markedly in appearance and location, the pituitary lesions being small and centrally situated whereas gliomas are larger and more irregular, yet the same prompt-driven procedure handles all of them without any class-specific adjustment. Finally, the *no tumor* case illustrates the asymmetry of the two branches described in Chapter 3: carrying no lesion, it receives no prompt and no mask is produced, so it is processed by the classification branch alone, which assigns it its category directly from the pooled image embedding.

4.6 Discussion

The two experimental chapters of this work address a single framework from two different angles: Chapter 3 established a MedSAM-based architecture for joint tumor segmentation and classification, and Chapter 4 examined a single component of that architecture, the classification head, replacing it with a variational quantum circuit. Read together rather than separately, the results say something more specific about where the value of this thesis lies than either chapter says on its own.

The strongest and least qualified result of this work is the segmentation performance reported in Chapter 3. A Dice coefficient of 0.9506 and an IoU of 0.9078 place the proposed framework above every retrained baseline evaluated under the same protocol, including attention-enhanced and Transformer-based architectures, and marginally above the published state of the art on BRISC. Crucially, this result does not depend on any single design choice examined in isolation: the ablation reported in Annex 4.7 shows that pretraining, fine-tuning and multi-bounding-box prompting each contribute a measurable and additive fraction of the total gain, and the zero-shot performance of MedSAM already exceeds several retrained baselines before any BRISC-specific adaptation is applied. This is a result of a different nature from those discussed later in this section: it is large, it is consistent across metrics, and part of it is attributable to a resource, large-scale medical pretraining, that the competing architectures simply did not have access to. The comparison with the published state of the art must nevertheless be read with the same caution expressed in Chapter 3: the proposed framework receives strictly more information than a fully automatic baseline, since it is prompted with bounding boxes, so the margin is better read as a demonstration that prompt-driven foundation models are competitive with automatic architectures under an equally rigorous protocol, rather than as an unqualified improvement.

The classification branch, by contrast, is a secondary result by design. It was introduced to demonstrate that a second task can be grafted onto the same frozen representation at negligible architectural cost, not to push classification accuracy to its ceiling. Its value in this thesis is structural rather than numerical: because the encoder is frozen and the head is the only trainable component standing between the shared representation and the decision, the classification branch of Chapter 3 constitutes exactly the controlled substrate that the experiments of Chapter 4 require. The MLP accuracy of 84.30% should therefore not be compared, on its own, against the classification-only literature reviewed in Chapter 2, where dedicated pipelines reach accuracies above 98% on similar tasks; those pipelines are trained specifically for classification, whereas the head evaluated here works with features optimized primarily for segmentation, under a backbone it cannot adapt. The relevant comparison is internal, between classification heads sharing that same constrained representation, which is precisely the comparison Chapter 4 conducts.

Within that internal comparison, three findings emerged, and their direction was remarkably consistent. The choice of data embedding produced by far the largest effect: angle embedding outperformed amplitude embedding by roughly 26 percentage points in accuracy, a gap far too large to be attributable to a single training run, and consistent with the mechanistic explanation available for it, namely the incompatibility of amplitude embedding with data re-uploading and the information loss imposed by its normalization step. The effect of qubit width was smaller but monotonic and consistent across all three classification metrics, eight qubits being both the widest configuration tested and the only one to exceed the classical MLP. The effect of circuit depth followed the same pattern: accuracy, Macro-F1 and balanced accuracy all improved monotonically from one to three layers, and the deepest configuration had in fact not fully converged when training stopped, suggesting that its reported result may understate what the configuration can ultimately reach.

The quantum head reaches the best result on all three classification metrics, and this outcome is not an isolated coincidence: it is the endpoint of a coherent sequence of ablations in which every design decision, from the embedding strategy to the qubit count to the circuit depth, pointed in the same direction. Angle embedding, eight qubits and three layers each independently improved performance over their alternatives, and the final configuration combining all three outperforms the classical MLP on accuracy, Macro-F1 and balanced accuracy simultaneously. This convergence across three independent metrics and three independent design axes is considerably more informative than any single comparison taken in isolation, and it is what justifies retaining the eight-qubit, three-layer configuration as the final architecture. The one caveat this thesis is careful to state, in keeping with the same experimental rigor applied throughout, is that the final margin over the MLP was obtained from a single training run per configuration; replication over several seeds, proposed as future work, would consolidate rather than test this result, since the ablations already point unanimously in its favour.

Alongside this performance, the quantum head achieves something the classical MLP cannot match under any circumstance: it does so with 2,708 trainable parameters against 33,924, a reduction of 92% that holds exactly, in every configuration tested, independently of any statistical qualification. This is the most robust finding of the quantum study, and it directly demonstrates the central promise of variational quantum circuits, namely the ability to represent a rich decision boundary within a drastically smaller parameter budget. The counterpart to this efficiency is a lower throughput on the classical simulator used for these experiments, 1.5 images per second against 6.0 for the MLP, a cost that stems from simulating the exponential state space of the circuit on conventional hardware rather than from any property of the circuit itself, and one that is expected to shrink substantially as quantum hardware and simulation methods mature.

Placed side by side, the two chapters make complementary contributions that reinforce a common thread. The segmentation result is the primary contribution of this thesis: it is large, corroborated by an ablation, and consistent with an established literature on the value of medical foundation-model pretraining. The quantum result is its natural extension: it demonstrates, under exactly the same rigor of controlled comparison, that a variational quantum circuit can match or exceed a classical classification head while using a fraction of its parameters, establishing quantum-enhanced classification as a credible and resource-efficient direction for this framework rather than a purely theoretical curiosity. Confirming the full extent of this advantage across multiple random seeds is the natural next step, and the ablations reported here provide a clear and consistent basis on which to pursue it.

What the two chapters share is a common commitment to controlled comparison. Every claim about the segmentation architecture in Chapter 3 rests on retraining every baseline under the identical protocol used for the proposed framework, and every claim about the quantum head in Chapter 4 rests on holding the backbone, the data and the training procedure fixed while varying a single component of the classification branch. This is what makes the two sets

of results interpretable together: the segmentation gains can be attributed to specific, isolated design choices rather than to an uncontrolled combination of factors, and the same discipline underlies the quantum ablations, whose effects, though smaller in magnitude, were consistent across every metric and every configuration examined. The main contribution of this thesis is consequently twofold: a segmentation framework that is demonstrably strong on its own terms, and a quantum classification component that matches its classical counterpart in performance while requiring a fraction of its parameters, establishing a compact and promising foundation for quantum-enhanced medical image classification.

4.7 Conclusion

This chapter investigated the integration of a variational quantum circuit into the classification head of the proposed MedSAM framework. Under a fixed and frozen backbone, three ablations isolated the effect of the data-encoding strategy, the number of qubits and the circuit depth, and angle embedding with eight qubits and three variational layers gave the best configuration among those tested, reaching an accuracy of 85.40% against 84.30% for the classical MLP while using 92% fewer trainable parameters. This last figure is the most robust result of the chapter, holding exactly across every configuration evaluated, whereas the accuracy gain remains within the range expected from a single-seed run. The quantum head is therefore best described as a compact and competitive alternative to the classical head, and confirming this advantage with multiple random seeds is left to future work.

General Conclusion

On the segmentation side, the proposed framework combined two-stage full fine-tuning of MedSAM with a multi-bounding-box prompting strategy designed to handle fragmented and irregularly shaped tumors. This approach reached a Dice coefficient of 95.06% and an IoU of 90.78% on the BRISC test set, outperforming every segmentation architecture retrained under an identical protocol, including attention-enhanced and Transformer-based baselines, and slightly exceeding the best published result on the benchmark. The controlled ablation conducted in Chapter 3 further showed that this performance is not the product of a single design choice but the additive result of medical pretraining, full fine-tuning and adaptive prompting acting together, which constitutes the central contribution of this thesis.

On the classification side, a lightweight MLP head applied to the same frozen representation reached an accuracy of 84.30%, establishing the reference against which the quantum classification head, developed and evaluated in Chapter 4, was compared. Through a systematic ablation of the data-embedding strategy, the number of qubits and the circuit depth, the eight-qubit configuration with angle embedding, data re-uploading and three variational layers emerged as the best-performing quantum architecture, reaching an accuracy of 85.40% together with higher Macro-F1 and balanced accuracy than the classical head, while using approximately 92% fewer trainable parameters. This parameter efficiency, observed consistently across every configuration tested, stands as the most robust outcome of the quantum study, and demonstrates that a circuit of only a few thousand parameters can meet or exceed the performance of a classical head more than ten times its size.

This work addressed the problem of brain tumor analysis from MRI images by developing a MedSAM-based framework for joint tumor segmentation and classification on the BRISC dataset. The study pursued two complementary objectives: adapting a medical foundation model to achieve accurate and reliable tumor segmentation, and investigating whether a variational quantum circuit could serve as a compact and competitive alternative to the classical classification head operating on the same shared representation. Taken together, the results above show that both objectives were met: the segmentation framework provides a substantial and well-corroborated improvement over existing approaches on BRISC, while the quantum classification head establishes, under the same standard of controlled experimentation, that quantum-enhanced components can be integrated into a medical foundation-model pipeline without sacrificing performance and at a markedly reduced parameter cost. The consistency of the classification gains across three independent design axes, embedding, width and depth, gives confidence in the direction of these findings.

Confirming the statistical significance of these findings with certainty would, however, require training over multiple random seeds, an extension left to future work; the results reported here rest on the consistency of the ablation trends rather than on a seed-level statistical test.

Several directions naturally follow from this work. On the technical side, evaluating the framework on additional brain tumor datasets and imaging modalities would clarify the generalization of both the segmentation and the quantum classification components beyond BRISC. Exploring deeper or more expressive quantum circuits, alternative entangling topologies, and eventually execution on physical quantum hardware would help determine how much of the parameter efficiency observed in simulation carries over to a real deployment setting. Extending the framework beyond single-modality MRI toward multimodal medical data, combining imaging with clinical records or physiological signals, would also bring the system closer to the conditions of real clinical practice.

More broadly, this thesis can be viewed as an initial step toward more autonomous and integrated medical AI systems. The combination of a promptable foundation model, a compact and efficient classification component, and a rigorous experimental methodology developed here provides a foundation on which future research, within the PRIME laboratory and beyond, could build agentic systems capable of coordinating multiple specialized models, reasoning over multimodal patient data, and supporting clinical decision-making in a transparent and reliable manner.

In summary, this work shows that a fully fine-tuned MedSAM framework, enhanced with an adaptive multi-bounding-box prompting strategy, can deliver state-of-the-art brain tumor segmentation, and that this same shared representation can support a quantum classification head that matches classical performance at a fraction of the parameter cost. It establishes a solid and extensible foundation for further research at the intersection of medical image analysis, vision foundation models, and quantum machine learning, and provides a meaningful starting point for continued graduate work aimed at building robust, efficient and clinically relevant AI systems.

Annex A – Complete Segmentation Results on BRISC 2025

Table 4.8: Segmentation results obtained on the BRISC 2025 test set, ordered by decreasing Dice coefficient. Higher values indicate better performance.

Rank	Model	Dice	IoU	Precision	Recall
1	**MedSAM full fine-tuning + multi-BBox (proposed)**	**0.9506**	**0.9078**	**0.9539**	**0.9491**
2	MedSAM full fine-tuning	0.9390	0.8875	0.9396	0.9408
3	MedSAM fine-tuning (multi-BBox)	0.9307	0.8735	0.9300	0.9345
4	MedSAM fine-tuning (global BBox)	0.9270	0.8687	0.9292	0.9292
5	MedSAM zero-shot	0.9005	0.8291	0.9227	–
6	SABERNet	0.8881	0.8123	0.9007	0.8874
7	TransSABERNet v2 (pretrained MiT-B2)	0.8848	0.8065	0.8910	0.8906
8	TransSABERNet v2 (140 epochs, Tversky)	0.8845	0.8063	0.8672	0.9132
9	SegFormer-UAttNet	0.8840	0.8067	0.8971	0.8822
10	U-Net	0.8801	0.7975	0.8952	0.8740
11	ABANet	0.8785	0.7982	0.8773	0.8950
12	ResU-Net	0.8764	0.7917	0.8903	0.8741
13	Attention U-Net	0.8730	0.7863	0.8818	0.8771
14	TransSABERNet	0.8718	0.7887	0.8811	0.8770
15	YOLO26l-seg (full fine-tuning, 150 epochs)	0.8596	0.7962	0.8857	0.8881
16	U-Net++	0.7970	0.6920	0.8078	0.8399
17	DeepLabV3+	0.7970	0.6920	0.8078	0.8399
18	TransResUNet	0.7773	0.6685	0.7973	0.7940
19	YOLO26l-seg (transfer learning, no fine-tuning)	0.7641	0.6959	0.7382	0.8163

All models were trained and evaluated on the same data splits, with the same augmentation and the same evaluation procedure. A dash indicates a metric that was not recorded for the corresponding configuration.

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